UNDERGRADUATE FINAL YEAR PROJECT REPORT

NED University of Engineering and Technology
Department of Electrical Engineering

Deep Learning Techniques Reviewal for Single-Step Wind Power Forecasting

Group Number: 32 **Batch: 2019-20**

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Author's Declaration

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Statement of Contributions

- Syed Hassan Ali Rizvi
  - Literature Review on Temporal Convolution Network
  - Data Acquisition
  - Implementation of Code
  - Report Writing
- Zeeshan Arif
  - Literature review on CEEMDAN
  - Data Acquisition
  - Data pre-processing
  - Implementation of Code
  - Report Writing
- M. Ameer Qureshi
  - Literature review for Transformer Model
  - Code Implementation
  - Report Writing
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  - Literature review for EMD and EEMD
  - Data Pre-Processing
  - Code Implementation
  - Report Writing

Executive Summary

The reliable and effective integration of wind energy into the electricity grid is the primary goal of wind power forecasting, which is undertaken for a number of significant reasons. Accurate estimates of the amount of electricity that can be produced by wind turbines over certain time periods, often ranging from a few minutes to several days in advance, are provided by wind power forecasting. The reliable and efficient operation of electricity systems depends on accurate wind power predictions. However, it is difficult to develop a wind forecasting model capable of reliably predicting such time-series due to the high randomness and non-stationary patterns of wind power time-series.

This report compares two deep learning techniques, Temporal Convolution Network (TCN) and Transformer Model, for Wind Power Forecasting. Additionally, decomposition techniques are used to preprocess the wind power data and reduce the influence of random noise on the prediction model. Both, Wind Power Forecasting and Deep Learning Models, are important tools that are used by wind farms to predict the output power. The models are tested against other cutting-edge methods after being trained using wind power information. In order to further preprocess the wind power data and lessen the impact of random noise on the prediction model, decomposition techniques EEMD and CEEMDAN are used. The results of the comparison between models showed Transformer model proved to be a better deep learning technique than TCN. Additionally, the integration of decomposition techniques with Transformer Model further strengthened the theory of transformer being the right model for wind power prediction. With this, this report would be suggesting the best technique to be used after the comparisons of the two deep learning models. Transformer Model is the preferred choice for Wind Power Forecasting relative to the research conducted in this project.

Acknowledgments

This project was meant to be the distinguished research in the field of wind power forecasting. Project revolved around least error generating technique but this research was not possible without the precious data provided by Sotavento Wind Farms, Turkey Wind Farms and other companies working for the betterment of this field with open hearts and constructive mindsets. Because they provided the best possible data with ideal time stamps and favorable conditions. Despite this, our project was not an easy task for the under graduate students with bare minimum knowledge of the deep learning but it was made easy by our prestigious teachers. Sir Ali Baig as Advisor and Sir Rashid Hussain as a Supervisor worked diligently for the whole year. Their advises, their suggestions and their motivation never let us go hopeless but always encouraged us to succeed. Nonetheless, we would thank our peer Hammad Ali Khan for helping us in the time of need with pure intentions and leading us out through the trouble. Lastly, we would thank our university for providing the best knowledge and best services which made this project possible and we wish to publish our research in international conferences.

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United Nations Sustainable Development Goals

The Sustainable Development Goals (SDGs) are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including poverty, inequality, climate change, environmental degradation, peace and justice. There is a total of 17 SDGs as mentioned below. Check the appropriate SDGs related to the project.

- ☐ No Poverty
- ☐ Zero Hunger
- ☑ Good Health and Well being
- ☑ Quality Education
- ☐ Gender Equality
- ☐ Clean Water and Sanitation
- ☑ Affordable and Clean Energy
- ☑ Decent Work and Economic Growth
- ☑ Industry, Innovation and Infrastructure
- ☐ Reduced Inequalities
- ☑ Sustainable Cities and Communities
- ☑ Responsible Consumption and Production
- ☑ Climate Action
- ☐ Life Below Water
- ☑ Life on Land
- ☐ Peace and Justice and Strong Institutions
- ☑ Partnerships to Achieve the Goals

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Following students have compiled the final year report on the topic given below for partial fulfillment of the requirement for Bachelor's degree in Electrical Engineering.

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1 Introduction

1.1 Background Information

Due to its potential to lessen greenhouse gas emissions and positive effects on the environment, wind energy is a renewable energy source that has attracted a lot of attention recently. The endeavor to switch to sustainable and cleaner energy sources has become more dependent on harnessing wind energy and turning it into electrical power [1]. The integration of wind energy into the electricity grid, however, faces substantial obstacles due to the fluctuation and uncertainty involved in its generation. Accurate and trustworthy estimates of wind energy production are necessary for the integration of wind power into the grid. Wind power forecasting is essential to the planning and operation of power networks, allowing grid operators to balance demand and balance supply while also assuring grid stability and reducing the need for expensive backup power sources. Because wind speed is erratic and heavily dependent on weather, projecting wind power is a challenging undertaking [2]. The reliability, security, and economical performance of a power system can all be enhanced by accurate wind power forecasting. The forecasting accuracy of wind power generation has been increased using deep learning algorithms. Deep learning models are continuously improved and updated with adjustments. Techniques such as Deep Convolutional Networks (DCN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and even attention-based models such as The Transformer, are being used to demonstrate their ability towards Wind Energy and power prediction [3]. For forecasting wind power, some recent studies have concentrated on machine learning and deep learning methods. For instance, an extreme gradient boosting stacking model based on long short-term memory (LSTM) neural networks, random forests, and has been presented [4]. Various other models that have been integrated with different topological models have also been used for Wind Power Forecasting, showing visible results.

1.2 Significance and Motivation

Due to its advantages for the environment and ability to lower greenhouse gas emissions, wind power is a renewable energy source that has attracted a lot of attention lately. The attempt to switch to sustainable and cleaner energy sources on a worldwide scale has made harnessing wind energy and transforming it into electrical power crucial. However, integrating wind energy into the electricity system will be extremely difficult due to the variability and uncertainty involved in its generation.

1.2.1 Bridging the Gap of Wind Power Integration

Accurate wind power forecasting is important because there is a pressing need to close the gap between the expanding demand for renewable energy and the difficulties involved in integrating it into current power networks. Wind power is an essential part of the sustainable energy mix as countries work to lessen their reliance on fossil fuels and minimize the effects of climate change [5]. Wind resources' inherent instability and unpredictability, however, prevents a seamless grid integration. This project seeks to play a crucial role in easing the smooth integration of wind energy, improving grid stability, and maintaining dependable power supply by developing enhanced wind power forecasting algorithms.

1.2.2 Optimization Resource Allocation and Grid Operations

1Accurate forecasts of wind energy generation over a range of time periods, from minutes to days in advance, are necessary for effectively utilizing wind energy resources [6]. Grid managers and energy planners can optimize resource allocation, scheduling, and dispatch of power producing units with the help of accurate forecasts [7]. Power grid operators can balance supply and demand, improve grid stability, and lessen wind power curtailment with the use of real-time wind power predictions, which has positive effects on the economy and environment [8].

1.2.3 Economic and Environmental Implications

Inaccurate wind power projections can result in poor power system decisions, supply and demand mismatches, and a greater reliance on expensive backup power sources, including fossil fuel-based power plants [9]. These inefficiencies have an effect on electricity pricing and can have a significant financial impact on utilities and consumers. Additionally, the efficient use of wind energy lessens the need for fossil fuels, which lowers greenhouse gas emissions and promotes a cleaner, more sustainable world [5]. The creation of cutting-edge deep learning methods for wind power forecasting has the potential to lessen these negative economic and environmental effects and promote a more robust and environmentally friendly energy sector [10].

1.2.4 Enhancing Energy Market Competitiveness

Energy market competitiveness is essential for fostering innovation and reducing costs in a sector that is continually changing [9]. By helping them to better manage their energy output and engage in the energy markets, accurate wind power forecasting gives renewable energy producers a competitive edge. Precise forecasts equate to lower risks and more trust in wind power projects for energy traders, investors, and stakeholders, supporting ongoing growth and expansion in the renewable energy sector [11].

1.2.5 Advancements in Deep Learning Techniques

This study makes contributions to the broader field of deep learning applications in addition to the subject of wind power forecasting. Deep learning models are faced with particular difficulties because of the complexity of wind data and the requirement for spatiotemporal representations [6]. It advances machine learning techniques and illuminates their potential applications in other renewable energy fields and beyond by investigating and optimizing several deep learning architectures for wind power forecasting.

1.3 Aims and Objectives

This report's main goal is to investigate and assess the usefulness of deep learning methods for wind power forecasting. The study aims to show the effectiveness of these cutting-edge machine learning models in tackling the problems brought on by the inherent fluctuation and uncertainty of wind energy through thorough analysis and experimentation. It will focus on two major deep learning techniques: Temporal Convolution Network (TCN) and an attention-based model called Transformer. Additionally, it will also be using decomposition techniques such as Ensemble Empirical Mode Decomposition (EEMD) as integration tool with the models. The report aims to provide a comparison study between these two models for wind power forecasting through formal analysis, research and experimentation. This project intends to contribute to the successful integration of renewable energy into existing power networks, fostering a sustainable and cleaner energy landscape by achieving more precise and dependable wind power forecasts.

The objectives of this report begin off by conducting a thorough literature research to examine the current forecasting methodologies for wind energy, including statistical methods and conventional numerical weather prediction models. Find the shortcomings and holes in the current forecasting methods, particularly with regard to long-term forecasts and handling intricate spatiotemporal correlations in wind data. This goal is to lay a solid foundation for the proposed study and find areas for improvement that deep learning techniques can address. With literature research for the report, obtaining pertinent wind data from trustworthy sources, such as historical wind speed and direction data, meteorological data, and related power generation data is implemented simultaneously. The preparation of a high-quality dataset that accurately reflects the temporal and spatial aspects of wind patterns is the goal of this mission. As we are focusing on two deep learning models, we will only be implementing and studying for these for our project. To ensure model generalization and resilience, use the proper validation approaches. This goal is to provide accurate and timely estimates of wind power generation by maximizing forecasting efficiency and accuracy of deep learning models. The goal is to implement our pre-processed data onto these models and to compare a solution of them once the model is trained. This will help us create a valid comparison paper for these techniques, with the right set of justifications.

This research paper's aims and objectives include a thorough investigation of the use of DL methods for efficient prediction of wind power. This study aims to add significant knowledge to the field of renewable energy research and offer workable solutions for sustainable energy generation and management by addressing the difficulties of wind energy integration, improving grid stability, and promoting a greener energy landscape.

1.4 Methodology

For this report, we are focusing on two deep learning models that are discussed below. These deep learning models were selected after evaluation of current research and experimentation of deep learning models for Wind Power Forecasting.

1.4.1 Temporal Convolution Network Implementation

The Temporal Convolution Network is a crucial part of our approach to forecasting wind energy. Convolutional neural networks of the sort known as TCN are very good at learning long-range temporal relationships, which makes them suitable for time series forecasting problems. Multiple layers of 1D convolutional filters make up the TCN architecture, which scans the temporal sequence to capture patterns at various time scales. Using historical meteorological conditions and electricity generation as input and output, the TCN model is trained on the preprocessed wind data. To reduce predicting errors, the model is optimized using suitable loss functions and gradient-based optimization techniques like Adam.

Additionally, two decomposition techniques: Ensemble Empirical Mode Decomposition (EEMD) and Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEDMAN) are integrated with the model to evaluate additional comparison results.

1.4.2 Transformer Model Implementation

The Transformer model is another essential component of our methodology, which has gained significant attention for its success in natural language processing tasks and sequence-to-sequence learning. The intricate spatiotemporal interactions in wind data can be successfully captured by the Transformer model in wind power forecasting.

The self-attention mechanisms in the Transformer model architecture allow it to model long-range interdependence without recurrent connections. The model uses numerous self-attention layers to assess the relative weights of various time steps, allowing it to zero in on pertinent temporal patterns. To manage the sequential nature of the wind data, the Transformer model additionally incorporates positional encoding and feed-forward neural networks. Similar to TCN, we train the model on the same set of data. This will enable us to provide a legitimate comparison between the two models.

1.4.3 Ensemble Empirical Mode Decomposition (EEMD) and Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEDMAN)

To improve the forecasting abilities, we use two decomposition techniques—EEMD and CEEDMAN—along with the TCN and Transformer models. While CEEDMAN enhances EEMD by using adaptive noise to retain data completeness, EEMD adaptively sifts the wind data into intrinsic mode functions (IMFs). For the TCN and Transformer models, additional input features include the decomposed IMFs and the residual component. The purpose of this inclusion is to better the forecasting capabilities of the models by capturing localized and non-stationary patterns in the wind data.

1.5 Report Outline

The scope of the project is a comparison study between deep learning models, TCN and Transformer, in conjunction with decomposition techniques such as Ensemble Empirical Mode Decomposition (EEMD) and Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEDMAN), for this project. The aim here is to not only evaluate on the results, but also address the current problems in the wind industry and its integration with power grids, and what are the possible solutions to it. The key sections of the report are as follows:

1.5.1 Chapter 2: Literature Review

This section will discuss wind power forecasting and deep learning methods superficially. It will focus on deep learning models, as well as describe about other models that are commonly seen in forecasting. Our two deep learning models will be introduced in this section, along with the decomposition techniques that are being integrated with them.

1.5.2 Chapter 3: TCN

This section will discuss the TCN model, all the pre-requisites required for TCN. This chapter will discuss about the basic fundamentals of the TCN model, along with any pre-requisites needed for model experimentation

1.5.3 Chapter 4: Transformer

This section would cover all the details regarding transformer technique and data processing.

1.5.4 Chapter 5: Decomposition Techniques

This section would cover all the details regarding transformer technique and data processing. This chapter would describe the working, analytics, model and limitations of 3 decomposition techniques namely EMD,EEMD and CEEMDAN.

1.5.5 Chapter 6: Model evaluation and Results

Results and data specifications would be provided.

1.5.6 Chapter 7: Conclusions

This portion would drive you to the summary of our project and recommendations regarding wind power.

2 Literature Review

2.1 Introduction

Due to the growing use of wind energy in the electric grid, wind power forecasting is a crucial tool in the energy sector. The dispatch of alternative generation sources and ancillary services can be preemptively informed by accurate forecasting, which can reduce risks and difficulties to the stability of the power system [12]. The fact that wind power time series are both highly volatile and non-stationary makes it difficult to model the raw time series using a single approach. With that in mind, numerous techniques have been used and implemented for wind power forecasting. These models include statistical, hybrid, deep learning and even numerical models. A study investigated various techniques for forecasting wind power, including the auto-regressive integrated moving average, the ordinary least squares method, and the k-nearest neighbors method [13]. Extreme learning machines (ELMs), recurrent neural networks (RNNs), support vector machines (SVMs), and artificial neural networks (ANNs) have all been found to be widely used for wind power forecasting, and the results have been found to be both promising and accurate [7]. Another study suggested an interval forecasting technique based on copula theory and long short-term memory (LSTM) networks to estimate the combined production of various wind farms [14].

Decomposition-based methods use decomposition techniques to divide the original time series into various sub-series that can be more accurately modelled than the original time series [15]. A mixed model prediction method for wind power time series based on EEMD decomposition was recently proposed in a study [15]. To further boost the precision of wind power forecasting, use EEMD to dissect the stochasticity of wind energy and investigate its regularity. A wind power forecasting system based on ensemble empirical mode decomposition (EEMD) and deep learning was proposed in another study [16]. Another study looked into how the effectiveness of several recurrent neural network-based forecasting models for short-term wind power generation was affected by empirical mode decomposition [12].

2.2 Traditional versus Deep Learning Models

Deep learning models and traditional machine learning models are two main groups of algorithms used in artificial intelligence and data science. Their architectures, techniques, and capabilities differ. The following is a comparison between traditional and deep learning models:

2.2.1 Architecture

Traditional Machine Learning: Traditional models are primarily based on statistical methods and necessitate manual feature engineering, in which domain specialists select appropriate data features to feed into the model. Traditional models include, among others, linear regression, decision trees, support vector machines (SVM), k-nearest neighbors (KNN), and Naive Bayes [17].

Deep Learning models are built on artificial neural networks that are inspired by the structure and operation of the human brain. These networks are made up of several layers of interconnected nodes (neurons) [18]. Deep learning models can learn hierarchical features automatically from raw data, eliminating the need for manual feature engineering.

2.2.2 Feature Development

Traditional Machine Learning: In traditional models, feature engineering is a critical stage. Engineers or domain experts must find and choose significant features from raw data, then preprocess and alter them so that they are suitable for the model [19]. Deep learning models may frequently learn valuable features from raw data, eliminating the need for extensive manual feature engineering. Deep learning is especially useful when working on the complex and high-dimensional data.

2.2.3 Requirements for Data

Traditional Machine Learning: When the dataset is small and the number of features is constrained, traditional models perform effectively. They may struggle with large amounts of data. Deep Learning: Deep learning models typically require vast volumes of data to learn complicated patterns effectively. They can handle high-dimensional data and frequently outperform large volumes of labelled training data.

2.2.4 Performance

Traditional Machine Learning: Traditional models are effective for a wide range of applications and can produce interpretable results. They are appropriate for issues with simple relationships between characteristics and the goal variable.

Deep Learning: Deep learning models excel in complicated pattern recognition tasks such as picture and speech recognition, natural language processing, and strategic game play. [20]They frequently outperform standard models in such scenarios, but because of their black-box character, they might be difficult to interpret.

2.3 Why do we need time series forecasting?

Time series forecasting is the fundamental need of any reliable generating unit. Time series analysis shows how data changes over time, and good forecasting can identify the direction in which the data is changing. Because of its ability to anticipate future values based on past data points in a time-ordered sequence, time series forecasting is a valuable tool in many professions and sectors. Here are some of the main reasons why time series forecasting is important [18]. Time series forecasting can assist firms in making educated decisions and planning for the future. Businesses may better allocate resources, manage inventory, optimize production, and establish realistic targets by forecasting future trends and patterns. Businesses can use time series forecasting to estimate future demand for their products and services, allowing them to change their supply chain and inventory management accordingly [8]. Time series forecasting is essential in meteorology for predicting weather patterns and climatic conditions. Time series forecasting aids in the efficient management of resources in areas such as energy, water, and transportation. Energy suppliers, for example, can utilize forecasting to predict electricity consumption and optimize power generation. Time series forecasting helps predict equipment breakdowns and maintenance requirements in manufacturing and industrial settings, lowering downtime and minimizing production disruptions.

Overall, time series forecasting is a helpful tool for anticipating future events and trends, making it an essential component of decision-making processes in a variety of disciplines.

2.4 Models used for Wind Power Forecasting

2.4.1 Physical Models

Physical models are the basic mathematical models that are based on different parameters that involve in a wind farm. In terms of wind power forecasting, a technique known as Numerical Weather Prediction (NWP) is used to produce forecasts for the entire region where a wind farm is located. NWP models typically require input data from a variety of sources, including weather observations, satellite imagery, and atmospheric soundings [21]. The output from the NWP is finally a forecast of weather conditions including wind speed and direction; the most important parameters for power forecasting

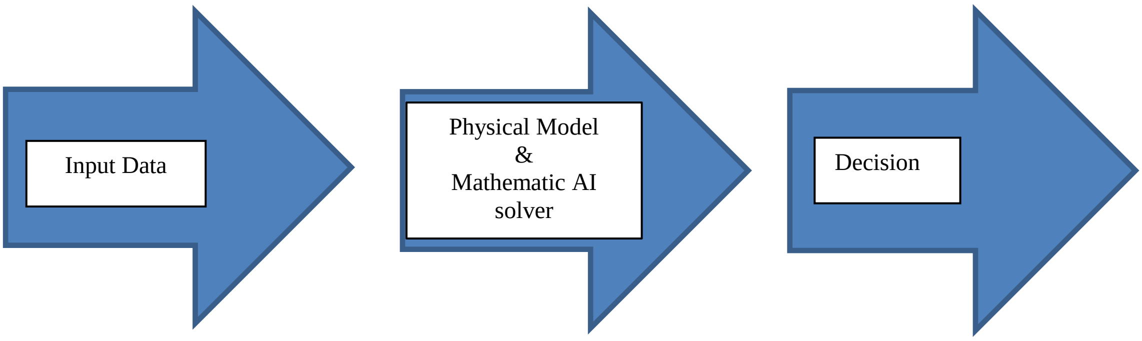


Figure 1: An overview of how physical models work

2.4.2 Statistical Models

These models use mathematics and statistics to describe relationships between variables. And when discussing wind power forecasting, these variables include wind speed, direction and power. Models such as ARIMA use historical data collected from wind farms and predict power. One of the key features of these statistical models is they use correlation, specifically autocorrelation, for analysis of data. Regardless of their mathematical uniqueness, they don't perform as expected during varied weather conditions [21].

2.4.3 Hybrid Models

Hybrid Models are those that combine different models together in order to improve wind power forecasting [22]. This model combination can include physical and deep learning models and even statistical and deep learning models. Overall, it uses the strengths of different models and combines them in order to solve the issue. A key factor is that, they have problems with stable prediction which can result in low training times, low efficiency [23].

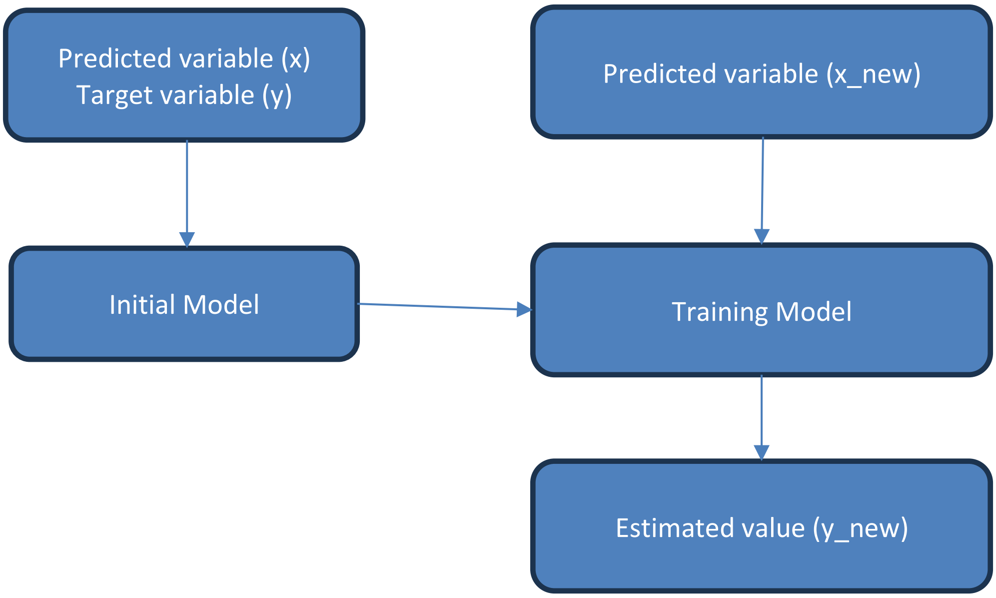


Figure 2: An overview of how hybrid models work

2.4.4 Deep Learning Models

Deep learning models are models that use convolution operation to extract of time series data and predict the output using classification results [24]. These models have a wide range of application and have been showing promising results in improving the accuracy in wind power prediction. Models such as CNN, LSTM and DBNs have provided fruitful results in time-series forecasting. This is due to a presence of multilayer neural networks for multiclass classification which exhibit superior results in wind power forecasting applications [18].

Table 1: A comparison of the different models used for wind power forecasting

Types of Models	Features	Cons
Physical	• Using numerical weather prediction (NWP) to predict weather. • Also includes wind farm factors and its geographical location.	• Large amount of weather dataset needed • The chances of inaccuracy seen in model is high.
Statistical	• Implementation of statistical models for wind power forecasting. • Models such as ARIMA and SVM are common examples used.	• Statistical methods don't handle weather data accurately.
Hybrid	• A combination of models, which can include deep learning, statistical and physical models.	• Low efficiency and high training times for these complex models is commonly seen.
Deep-Learning Model	• Extraction of time-series data features using convolution layers. • No need for extra data pre-processing.	• The quality of data input is a factor when considering the overall performance of the DL model.

2.5 Why do we prefer Deep Learning Models in Wind Power Forecasting?

All four models mentioned above have their pros and cons, but what makes deep learning models to be chosen from the lot is due to its ability to handle complex, long data [25]. By complex we mean the data is having multiple variables and is lengthier as well. Apart from this, the ability to capture dependencies, both long and short term, over time is one of the key features of it [26]. Models such as LSTM or even LSTM based RNN are able to achieve this.

The ability to handle nonlinear relationships is another quality possessed by deep learning models. These qualities help the models improve their efficiency by modelling these non-linear relationships between the variables. Another key feature of a deep learning model is adaptive; adapting to any changing conditions, either form new data or use existing data by adjusting their prediction in real-time. While physical models such as NWP do not respond well to weather changes which makes them less efficient as compared to deep learning models

In our case of wind power forecasting, we require a model that is accurate and reliable; a model that is efficient, whose computation time is less. Statistical, physical models don’t fit the criteria. Hybrid models can prove to be a well-suited option in certain conditions but then again, the complexity increases and the type of data required for these models limit the overall efficiency of the model and hence, the output would not be as expected.

2.6 TCN

The nature of the short-term forecast for wind power should be emphasized before adopting the TCN. Meteorological variables and previous wind speeds serve as the model's input and output variables, respectively. The main restriction is that wind power must be predicted by previously measured input factors for the upcoming hour, day, or year. TCN is suggested as a way to utilize parallelism to speed up training and assessment while also avoiding the issue of disappearing gradients. It is divided into three sections: residual connections, dilated convolutions, and causal convolutions. But we deliberately ignored the remaining link. This paradigm will be further discussed in Chapter 3.

2.7 Transformer

The Transformer model is a deep learning architecture that was introduced in the paper [27]. It revolutionized the field of natural language processing (NLP) and has since been widely used in a variety of machine learning problems other than NLP. By utilizing a revolutionary self-attention mechanism, the Transformer architecture tackles the constraints of earlier sequence-to-sequence models.

The Transformer design has served as the foundation for many cutting-edge NLP models, including BERT, GPT, T5, and others. It allows for parallelization, which speeds up training, and its attention method efficiently captures long-term dependencies. This adaptability has led to its use in fields other than sequence-based applications, such as image production and time series prediction. The model introduces a concept of self-attention; a new improvement made towards the natural language processing, which allows the inputs to be focused/attended accordingly and create the appropriate output image out of it. An in-depth explanation about the transformer model is on Chapter 4.

2.8 Decomposition Techniques

Decomposition in deep learning refers to the process of breaking down difficult tasks or models into simpler, more manageable components. Understanding, optimization, and modularity of deep learning systems are all aided by the decomposition process. Decomposition techniques are commonly utilized to address a variety of issues related to large-scale and complex deep learning models. Different Decomposition techniques can be used as implementation or supporting tool for wind power forecasting. In our case, we will be focusing on two major decomposition techniques: Ensemble Empirical Mode Decomposition (EEMD) and Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN). Both of the mentioned models are a subsidiary of the Empirical Mode Decomposition (EMD).

2.8.1 Empirical Mode Decomposition

It has been developed a novel method for assessing nonlinear and non-stationary data. The approach's fundamental component is the 'empirical mode decomposition' method, which allows any complicated data set to be decomposed into a limited and typically small number of 'intrinsic mode functions’. This decomposition method is adaptive and thus very efficient. Because the decomposition is based on the data's local characteristic time scale, it can be used to nonlinear and non-stationary processes. The main conceptual innovations in this method are the introduction of 'intrinsic mode functions' based on local signal properties, which make the instantaneous frequency meaningful; and the introduction of instantaneous frequencies for complicated data sets, which eliminates the need for spurious harmonics to represent nonlinear and non-stationary signals.

Unlike almost all previous data analysis approaches, the EMD method is adaptive, with the breakdown dependent on and derived from the data. The data x(t) is decomposed in terms of IMFs, c_j in the EMD technique;

$$x(t) = \sum_{j=1}^{n} cj + rn \tag{1}$$

where, r_n is the remnant of data x(t) after extracting n-IMFs.

2.8.2 Ensemble Empirical Mode Decomposition

EMD is a technique for breaking down a given signal into a series of Intrinsic Mode Functions (IMFs) or components. IMFs are functions that satisfy two conditions: (1) they have the same number of zero-crossings, extrema, or minima, and (2) the mean value of the envelope formed by the local maxima and the envelope defined by the local minima is zero at any point.

Ensemble Empirical Mode Decomposition (EEMD) is a data-driven signal processing technique for breaking down complex signals into their fundamental oscillatory components. It is an extension of the Empirical Mode Decomposition (EMD) approach that adds an ensemble averaging step to improve its performance and resilience.

While EMD is a powerful method for signal decomposition, it can occasionally suffer from mode mixing, which occurs when some IMFs have several oscillatory patterns, making it difficult to comprehend and analyze the signal properly. This restriction is addressed by EEMD, which introduces an ensemble of noise-assisted decompositions. The procedure of EEMD utilization is as follows:

1. **Generate white noise:** To accomplish EEMD, white noise is introduced to the original signal to create different signal realizations. This step aids in breaking up symmetries and fixing mode mixing problems.
2. **Perform EMD:** EMD is used to divide the signal into IMFs for each realization (original signal + additional noise).

3. **Averaging ensembles:** After applying EMD to all realizations, the related IMFs from each realization are averaged to generate the final IMF components.
4. **Residual components:** The residual component, which is the signal that remains after deleting all of the IMFs, is also obtained.

EEMD has various advantages over classic EMD, including better separation of oscillatory components, less mode mixing, and increased noise robustness. EEMD provides a more stable and accurate decomposition of complicated signals by using ensemble averaging, making it useful in a variety of applications such as signal processing, time-series analysis, and feature extraction in machine learning.

It is worth mentioning that EEMD is computationally more expensive than EMD because it requires the generation of several realizations and the application of EMD to each of them. However, the advantages of enhanced decomposition quality frequently outweigh the additional processing cost, particularly when working with difficult and noisy inputs.

2.8.3 CEEMDAN

Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) is a more sophisticated and improved version of Ensemble Empirical Mode Decomposition (EEMD) for signal processing and time-series analysis. It was proposed to alleviate some of EEMD's drawbacks and improve complicated signal decomposition.

CEEMDAN is a data-driven signal processing technique similar to EEMD that tries to breakdown a given signal into its intrinsic mode functions (IMFs). IMFs are signal components that represent various oscillatory patterns. CEEMDAN uses an adjustable noise amplitude to generate numerous signal realizations and performs the Empirical Mode Decomposition (EMD) procedure to each realization. Here is the general overview of the CEEMDAN process:

1. **Add adaptive noise:** CEEMDAN introduces an adaptive noise amplitude that fluctuates during the decomposition process. This adaptive noise is more effective than the fixed noise employed in EEMD in dealing with mode mixing difficulties.
2. **Generate realizations:** Realizations are generated by combining the original signal with adaptive noise, resulting in distinct signal realizations.

3. **Perform EMD:** EMD is applied to each realization in order to break it into Intrinsic Mode Functions (IMFs).
4. **Averaging ensemble:** Similar to EEMD, the IMFs from each realization are averaged to generate the final IMF components.
5. **Residual component:** The residual component, as in EEMD, is computed as the leftover signal after eliminating all IMFs.

CEEMDAN's variable noise amplitude aids in better isolating oscillatory components and decreasing mode mixing concerns, resulting in a more robust and accurate decomposition technique, particularly for complex and noisy signals.

CEEMDAN has been used in a variety of applications, including signal processing, finance, climate analysis, and biomedical signal analysis. It is regarded as an efficient method for extracting relevant information from time-series data, with promising results in a variety of practical applications.

3 Temporal Convolution Network

3.1 Introduction

Wind power variation and intermittency pose significant problems to distribution network operation and control. Accurate short-term forecasting for wind power is beneficial in avoiding risks associated with wind power uncertainties. The temporal convolutional network (TCN) is proposed in this research to increase the accuracy of short-term prediction for wind power. The suggested strategy, using dilated causal convolutions and residual connections, tackles the problem of long-term dependencies and performance degradation in deep convolutional models in sequence prediction. The simulation findings reveal that the TCN training process is quite stable and has a high degree of generalization [27]. Furthermore, TCN outperforms existing predictors such as the support vector machine, multi-layer perceptron, long short-term memory network, and gated recurrent unit network. Temporal convolutional network (TCN) is a deep learning model which is also known as dilated causal convolutional network [28]. The pre-existing deep learning models such as RNN, LSTM, GRU etc is quite limited in terms of long term dependencies. the two possible errors which occur in these models are exploding gradient and vanishing gradient [29]. due to these errors a typical convolutional neural network is employed to perform timeseries forecasting for our desired task. A temporal convolutional network (TCN) stands as a cutting-edge evolution of the one-dimensional convolutional neural network. This innovative approach enhances its capabilities by leveraging a combination of causal dilated convolutions and residual connections, resulting in an expansive receptive field. Remarkably, the TCN model has consistently exhibited superior performance when compared to traditional recur [30] rent neural networks (RNNs) and convolutional neural networks (CNNs) across diverse tasks, highlighting its efficacy in areas like action segmentation and network anomaly detection [31].

This remarkable superiority extends across a multitude of domains, from traffic prediction to audio processing, from machine translation to human motion detection. In each of these fields, TCN has demonstrated its prowess, consistently outperforming the widely recognized Long Short-Term Memory (LSTM) model, which has been a staple in sequential data analysis.

Imagine TCNs as a bridge between the power of convolutional neural networks in capturing spatial features and the sequential insight of recurrent neural networks. The introduction of causal dilated convolutions allows the network to see farther into the temporal domain, capturing long-range dependencies that are essential for understanding the patterns and relationships within time-series data. These dilated convolutions, in combination with residual connections, not only enhance the model's ability to capture essential temporal features but also alleviate the vanishing gradient problem often faced by traditional RNNs.

The growing adoption of TCNs signifies their versatility and efficacy, particularly in tasks where both local and global temporal relationships play a crucial role. This network architecture offers a fresh perspective on how to tackle time-dependent challenges, and as the field of deep learning continues to evolve, TCNs hold a significant place in the forefront of cutting-edge methodologies [15]. Their ability to surpass established models like LSTM in various applications underscores the significance of TCNs as a powerful tool in the data scientist's toolkit, promising to unlock new insights and push the boundaries of what's achievable in the realm of sequential data analysis [27].

A typical convolutional network is not suitable to cater the properties of time series data, a time series data is said to be a dataset in which each reading has been taken at a certain time stamps. Temperature and sound are quite examples of timeseries data. In time series data the two adjacent readings are correlated to each other, ie:- let’s say a data set of *n* entries in which we are currently at the index *i* the reading at this particular is quite correlated with the preceding one *i-1*, this Auto-correlation determines the tendency or degree of time series data.The proposed design of temporal convolutional network can be achieved by stacking multiple layers of dilated causal CNNs and ResNets for long term dependencies. The following details explain the properties of Causality & dilation

3.2 Causal Convolutions

The TCN is distinguished by two features: (1) the convolutions are causal, i.e., the current output is related exclusively to the present and past inputs and not to the future inputs; and (2) the convolutions are asymmetric. (2) Just like an RNN, architecture may accept a time series of any length and map it to output data of the same length. The initial layer of the TCN is a 1-dimensional fully convolutional network, which satisfies the first property. Each intermediate layer has the same size as the input layer, and succeeding layers get zero size paddings in order to maintain this size equality. TCN solely convolves the input from the present time and prior time in order to meet the second characteristic..

3.2.1 Causality

Causality can be simply defined as the property of a system or a network by which it can only be dependent upon present & the past values. A causal system is completely independent of the future values of the data as shown in fig below. In this way a traditional convolutional neural network is relevant to only the past values of the data and it becomes completely irrelevant to the future or upcoming values of the data.

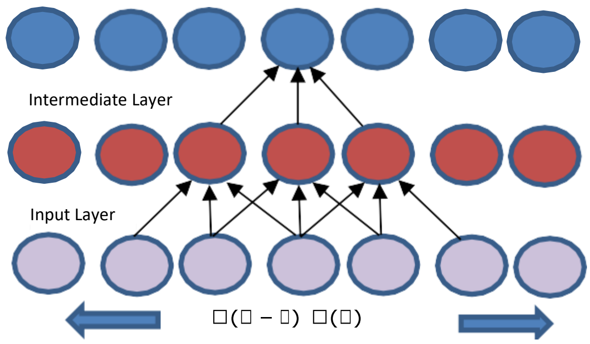


Figure 3: Causal Convolutions.

3.3 Dilated Convolutions

TCN decides to use dilated convolutions, which permit an exponentially huge receptive field, to apply causal convolution on time series with a lengthy history. In order to enhance the size of time series viewed by the network while maintaining essentially the same amount of computation, dilated convolution, as illustrated in the following image, adds certain weights to the convolution kernel while leaving the input data same. Formally, the operation F of dilated convolution on elements of the time series is defined as follows with a 1-dimensional time series X and a filter f:

$$F(s) = (x \times f)(s) = \sum_{i=0}^{k-1} f(i) \times x_{s-di} \qquad (2)$$

Where,

$*$ is the convolutional operation

k is the filter size

d is the dilation factor

s − di is the direction of the past

Therefore, using a set step size between each pair of adjacent filter taps is equivalent to using a dilation. The enlarged convolution becomes a normal convolution at d = 1. The output of the top layer may reflect a wider variety of inputs with the bigger expansion, thus enlarging the receptive field of TCN. TCN may broaden the receptive field in two different ways: 1) by raising the dilation factor d, and 2) by selecting bigger filter sizes k.

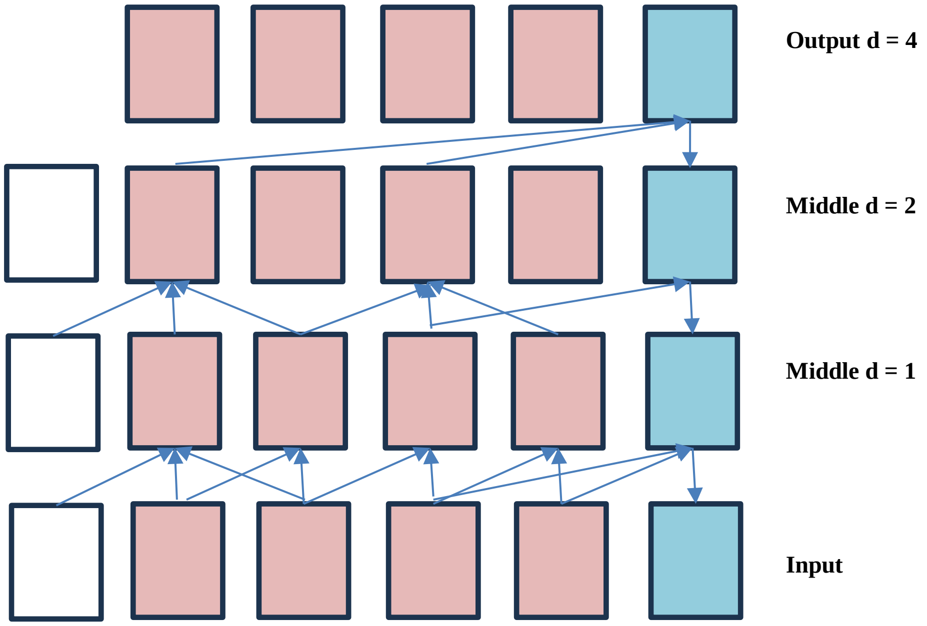


Figure 4: Dilation Process

3.4 Receptive Field

In deep learning models, a receptive field is ultimate field space which a final layered neuron caters. The term "receptive field" refers to the size of the input sequence that a convolutional layer can "see" or "attend to" when making predictions at a particular time step. In other words, it's the range of past inputs that the network considers when making predictions for a given time step. In short it represents the memory of the network. The long term dependencies of TCN is a function to the size of the receptive field, the bigger the size of the receptive field the greater memory the network constrained to the relevance of past data(correlation). One way to increase the size of receptive field in a TCN is to use a huge sized kernel of the convolutional filters used in the network. It is a simple and straightforward solution for long term dependencies of the network. But the above solution is computationally uneconomical as it wastes a lot of computational resources since the size of kernel matrix makes the weighs and biases calculation quite complex while the time required by a typical GPU is also another factor to drop this simple solution of large receptive field.

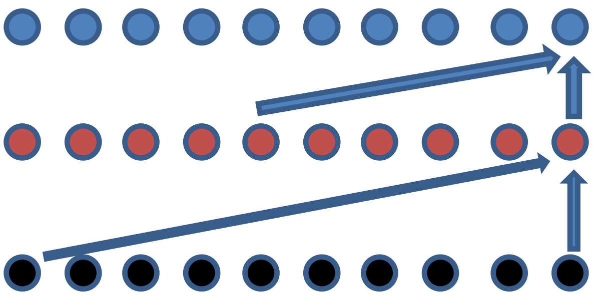


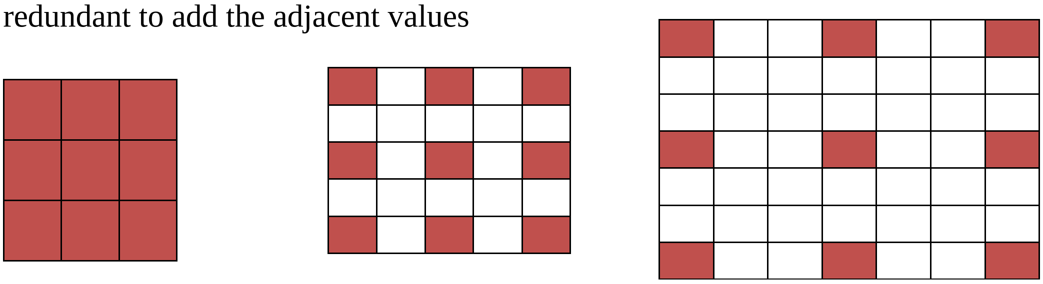
Figure 5: Receptive Field

3.5 Dilation

Dilation, also known as "atrous convolution," is a key concept in Temporal Convolutional Networks (TCNs) that allows the network to capture larger receptive fields while maintaining a compact model architecture. Dilation is particularly useful for processing sequential data with long-range dependencies, such as time series.

In traditional convolutional layers, each filter (or kernel) operates over a fixed-sized window of input values. For example, a standard 1D convolutional layer with a kernel size of 3 would consider a local window of three adjacent values from the input sequence at a time. This local window moves across the input sequence, and the convolutional filter learns to extract features from these local windows.

Dilation can be achieved by introducing holes or null values in the kernel matrix as shown in figure below, due to which the matrix size is increased but the calculation complexity remains same for that small kernel. The fact that we introduce holes in adjacent entries of kernel matrix is due to the nature of time series data. The adjacent readings of data in the time series dataset are quite similar and hence, it would be redundant to add the adjacent values



3x3 kernel matrix dilation rate=1

3x3 kernel matrix dilation rate=2

3x3 kernel matrix dilation rate=3

The dilation rate determines the spacing between the values in the filter, effectively increasing the receptive field of the filter without increasing the number of parameters or the computational cost as dramatically as a larger kernel size would. In Standard Convolution a traditional convolutional layer, each value in the filter corresponds to an adjacent value in the input. For example, with a kernel size of 3, the filter looks at three consecutive values in the input (X_1, X_2 and X_3 etc.)

While in a dilated convolution, the filter values are spaced out by a certain dilation rate, resulting in a larger receptive field. The filter "jumps" over a number of positions based on the dilation rate, allowing it to capture distant dependencies. By increasing the dilation rate, you can control how much the receptive field expands while keeping the computational cost manageable. This is particularly advantageous for processing long sequences with large temporal contexts, as TCNs can capture important patterns over extended time intervals. In TCNs, multiple dilated convolutional layers with increasing dilation rates can be stacked, allowing the network to capture a wide range of temporal dependencies and patterns in the input sequence without requiring an excessively deep architecture. This makes TCNs effective for tasks involving sequential data, such as time series forecasting, natural language processing, and audio processing

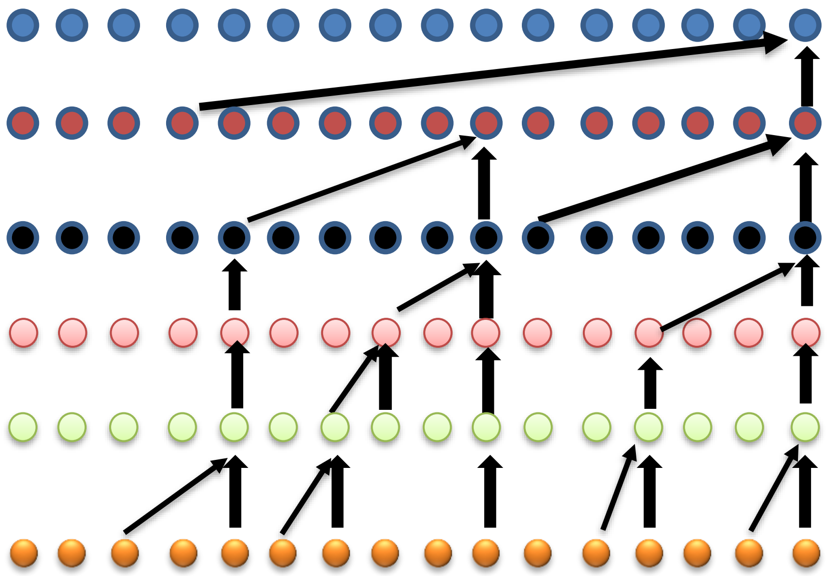


Figure 6: Detailed Receptive Field diagram

3.6 TCN Model

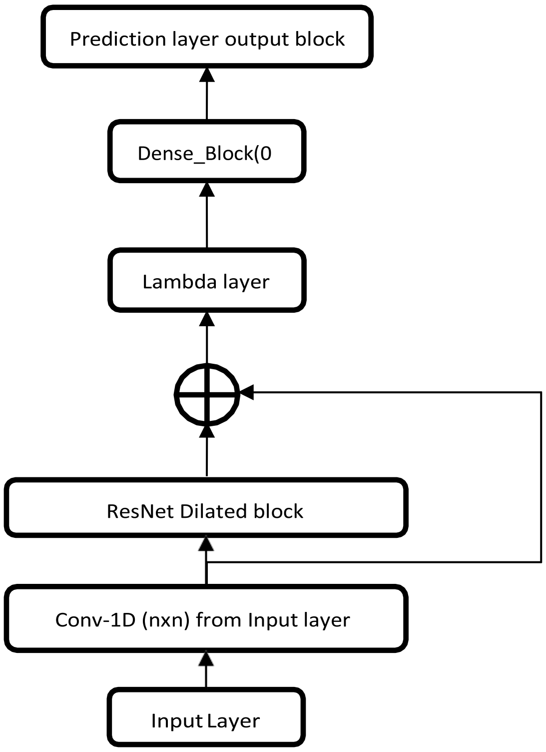


Figure 7: TCN Model Architecture.

4 Application of Decomposition Techniques on Wind Power Forecasting

4.1 Ensemble Empirical Mode Decomposition

4.1.1 Introduction

4.1.2

To address scale separation sans subjective intermittence test, Ensemble EMD (EEMD) suggests true IMF as mean trials, each with signal and finite white noise. 'Single' substitutes 'data' for whole data's decomposition, not specific part's identification. EEMD applies white noise's statistical features, viewing EMD as adaptive dyadic filter banks. Uniform mixed white noise aids signals' projection onto scales, refining via ensemble averaging. This surpasses EMD with noise-assisted data analysis. Ensemble mean, the sole constant, emerges with multiple trials. Finite white noise facilitates meaningful results within dyadic filter banks. EEMD's meaningful response is the noise-added signal ensemble mean of numerous trials.

4.1.3 Process of Ensemble Empirical Mode Decomposition

All data comprise signal and noise blends. The ensemble mean boosts measurement precision, using data from varied observations with diverse noise [36]. To extend this ensemble idea, noise is added to a singular dataset, X(t), simulating multiple experiment runs. The extra white noise reflects random measurement noise. The mth observation in this procedure becomes:

$$X_m(t) = X(t) + W_m(t) \tag{3}$$

As depicted in the equation, each instance the fresh ensemble incorporates new white noise element Wm(t). Despite noise inclusion diminishing signal-to-noise ratio, added white noise offers even reference scale distribution for EMD facilitation. Hence, low signal-to-noise ratio doesn't hinder but enhances decomposition, preventing mode blending [35]. Building on this notion, an extra stride is taken by asserting that infusing white noise could assist in extracting actual data signals. This is termed EEMD, a true NADA approach. Flowchart of EEMD

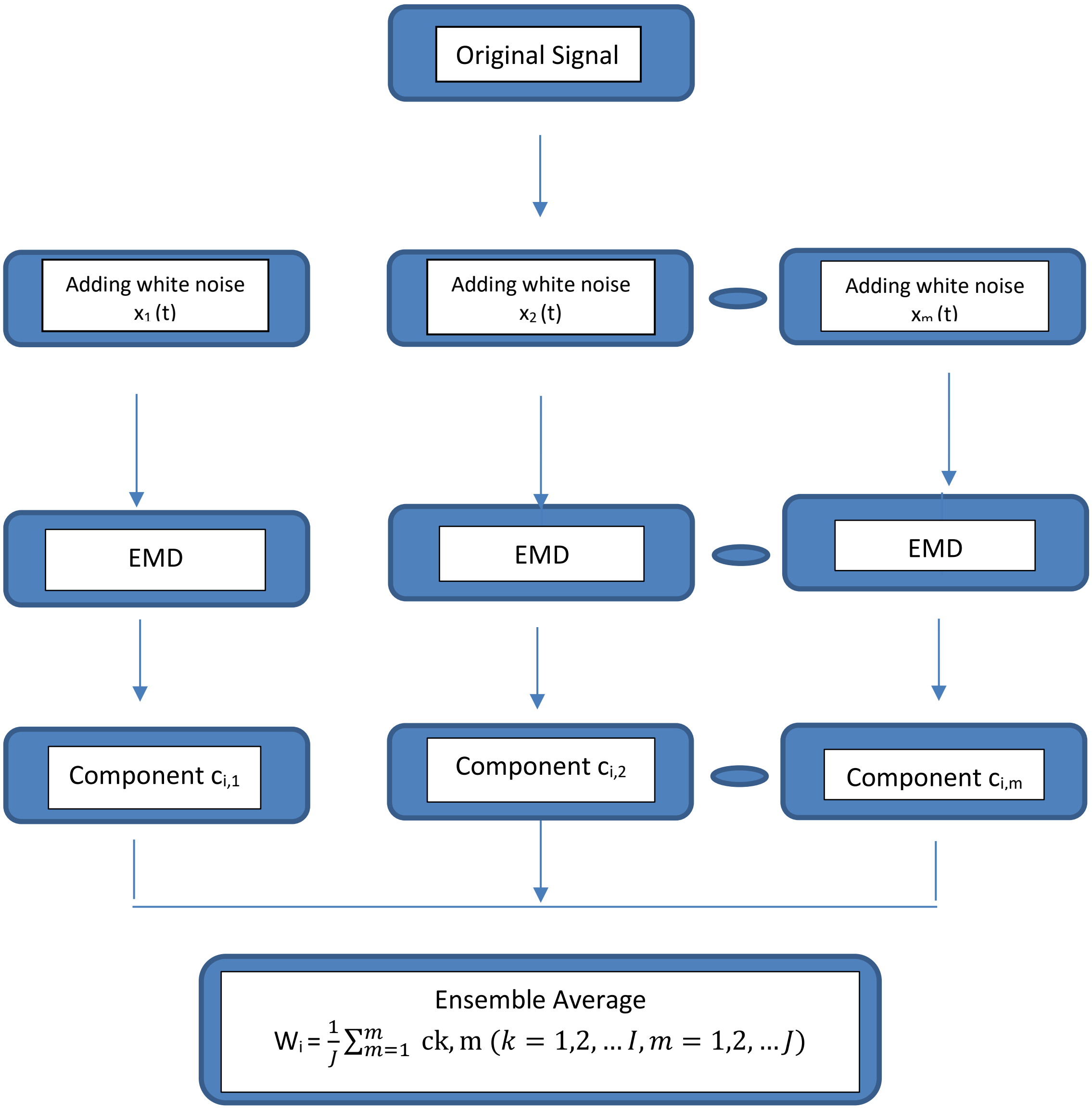


Figure 8: Flowchart showing process of EEMD.

4.1.4 Limitations of EEMD

1. **<u>Amplitude of Noise added</u>**

   If the added noise's magnitude is too slight, it might not introduce the extrema alteration crucial for EMD, particularly when data has steep changes. Consequently, for EEMD success, the added noise's amplitude needs amplification—avoiding insignificance. However, elevating ensemble count magnifies this effect, allowing reduction in added white noise to nearly negligible. In practice, a few hundred ensemble counts yield excellent results, and any remaining noise incurs <1% error if its amplitude is a fraction of original data's standard deviation [34].

2. **Post EEMD process aliasing**
This is the condition when EEMD IMF cannot be utilized by the next EEMD cycle due to mode mixing and aliasing of the previous IMFs. For this purpose, we introduce inter-procedural EMD to break the IMF in components by sifting method and avoid the frequency matching. Once EMD is conducted, then the EEMD processed IMFs start the summation and noise addition into the summated IMFs [37].

4.2 Complete Ensemble Empirical Mode Decomposition with Adaptive Noise

4.2.1 Introduction

As mentioned above, EEMD has issues with high computational costs and residual noise. As the number of trials grows, so does the number of sifting processes, and different realizations of signal plus noise produce a varying number of modes, making ultimate averaging problematic. A CEEMDAN was designed to reduce the number of trials while maintaining the ability to handle the mode mixing problem.

4.2.2 Process of CEEMDAN

CEEMDAN Algorithm provides an improved methodology for the removal of errors generated in EMD and EEMD. These are the following steps [20]:

1. Add white noise with zero mean value and unit variance to the original signal, and solve for the first residue r_1.

$$x_k = x + \beta E_{k+1}(w_i) \tag{4}$$

$$r_{k+1} = (M(x_k)) \tag{5}$$

where,

E_{k+1} is defined as the operator to generate the k^{th} mode obtained by EMD.

M is the operator to generate the local mean value of the applied signal.

x is the signal to be decomposed

w_i (constant) is the realization parameter of White Gaussian Noise with zero mean and unit variance

β_0 is the White Gaussian Noise weight coefficient.

2. Calculate the first mode D_1

$$D1 = x - r1 \tag{6}$$

3. Calculate the second mode D_2

$$r2 = (M(r1 + \beta1\, E2\, (wi\,))) \tag{7}$$

$$D2 = r1 - r2 \tag{8}$$

4. To calculate the k^{th} residue from k=1,2,…,I

$$R_k = (M\big(r_{k-1} + \beta_{k-1}E_k(w_i)\big) \tag{9}$$

5. Calculate the k^{th} order mode

$$D_k = r_{k-1} + r_k \tag{10}$$

6. Repeat from step 4 for further modes

4.2.3 Flowchart of CEEMDAN:

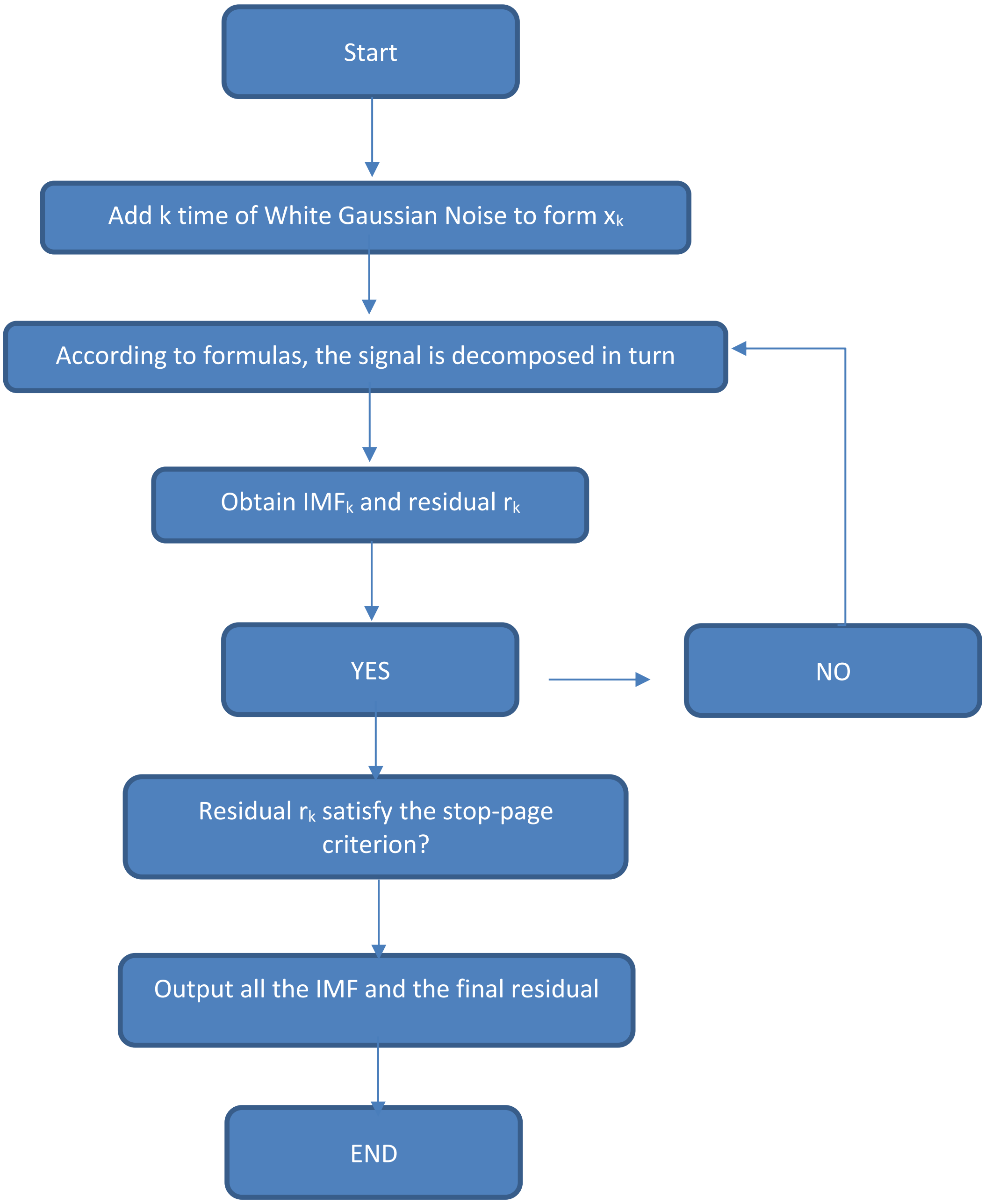


Figure 9: Flowchart showing the process overwiew of CEEMDAN

4.2.4 Limitations of CEEMDAN

CEEMDAN being an efficient tool for the compensation of EEMD and EMD, still may face some disorders which might need further filtration. Below mentioned are the issues [33] [35]:

1. Some part of residual noise is still included in the mode.

As the residue and IMFs are used in CEEMDAN, so we may face the overlapping of noise frequency with the required residue. This addition causes the erroneous data to be fed and cascaded.

2. Signal information appears later than EEMD. Sometimes, “False” patterns appear in the early stage of decompositions. This happens because of the erroneous signals passed on from the EMD stage. Usually it is the because of aliasing and mode mixing which is processed in EEMD and same thing appears in CEEMDAN. Thus, patterns are expected to be carrying the errors resulting in false metrics and graphs.

5 Transformer Model

5.1 Introduction

The transformer is a neural network architecture which has a foundation based on the mechanism of attention. It is a different model if compared with contemporary models such as CNN or RNN [3]. It is originally made as a natural language processing model, for tasks such as text generation, paraphrasing, sentence structuring and many more. The ability of this model to perform tasks completely different from the natural language processing model, is the addition of certain blocks, parameters and techniques within the model. These additions and improvisations allow the model to work as a prediction model, which we can then use in areas such as Wind Power Forecasting. The feed-forward network, self-attention mechanism, and residual structure make up the transformer network's fundamental unit. The network's ability to fully describe sequential data is made possible by the overall multilayer encoder to decoder topology [32]. The intricate spatiotemporal interactions in wind data can be successfully captured by the Transformer model in wind power forecasting. To manage the sequential nature of the wind data, the Transformer model additionally incorporates positional encoding and feed-forward neural networks.

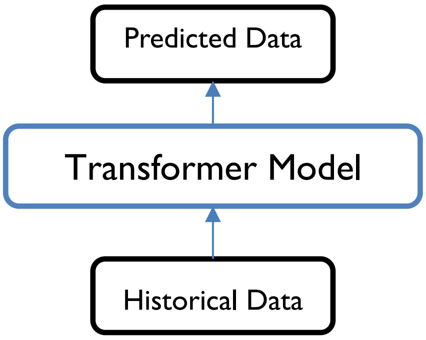


Figure 10: A basic overview of the Transformer Model

5.2 Model Architecture

Many researchers have implemented the use of an encoder-decoder model for power forecasting [4]. Studies have also shown the implementation of attention-based gated recurrent unit (AGRU) for forecasting which has shown positive results [33]. The encoder part of the transformer is responsible for taking in an input sequence, in the state called tokens, and producing a sequence of hidden features that capture information about the input sequence. The input sequence is converted into a fixed vector representation as $D = \{D_1, D_2, D_3, D_4 \ldots .D_n\}$. A self-defined number of identical encoder layers are placed one on top of the other to make up an **encoder**. The multi-head self-attention mechanism and the position-wise completely connected feed-forward network are the two sub-layers that make up each encoder layer. The output data is layer-normalized after each sub-layer employs a residual structure.

The **decoder** mechanism in the transformer model is responsible for generating the output sequence from the encoded input sequence. It gives the output in the same form as the input, i.e. $O = \{O_1, O_2, O_3, O_4 \ldots .O_n\}$. In our case, the output will be predicted wind power. Hence, we can show our output as $P = \{P_1, P_2, P_3, P_4 \ldots .P_n\}$. As shown in Figure, the transformer model includes other blocks as well such as the self-attention mechanism, point-wise fully connected layers and the ResNet structure that allow the encoder and decoder blocks to function correctly [34]. The embedding layer and all of the model's outputs have the same self-defined dimension d_{model} in order to promote residual connection. The encoder and decoder both have the same amount of stack layers. In our proposed model, Three sub-layers make up each decoder layer, which automatically makes the number of encoder layers to match up. The Masked Multi-Head Attention Layer, the first sub-layer, has the primary responsibility of ensuring that the forecasting of Position i only relies on the known outputs of Positions smaller than i. The encoder layer's final two layers employ the identical sub-layers. The output is normalized by layer and has a residual architecture for each sub-layer.

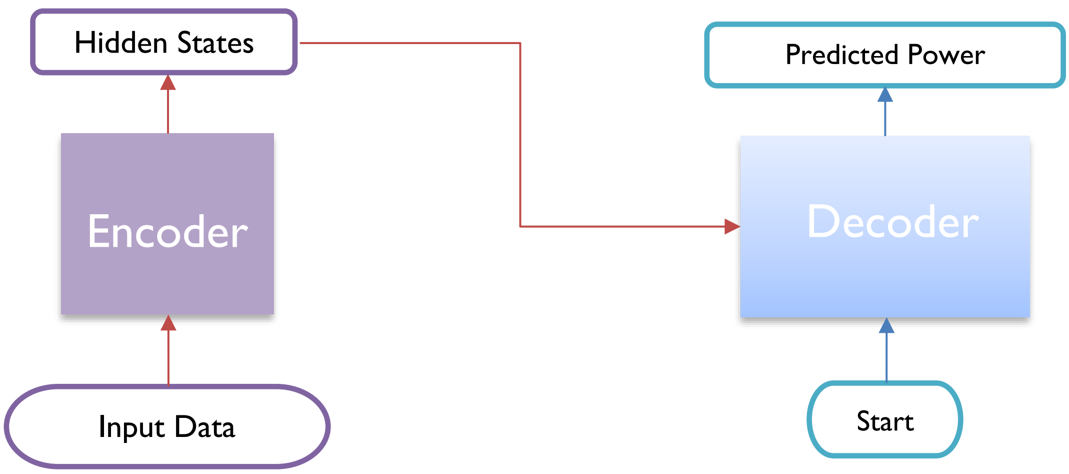


Figure 11: Encoder-Decoder mechanism of the Transformer Model

5.3 Attention Mechanism of the Transformer

Initially, the Transformer utilizes self-Attention to fully simulate input-output relationships. Applying the Transformer model to predict wind power time series enhances factors' grasp for improved forecasting results [33]. Drawing from human brain concepts, Attention focuses on varying elements in different contexts, diminishing or disregarding attention to others. An attention function connects vectors—query, key-value sets, and output—where the query's alignment with each key determines value contributions' weight. The outcome emerges as a weighted summation of values [35].

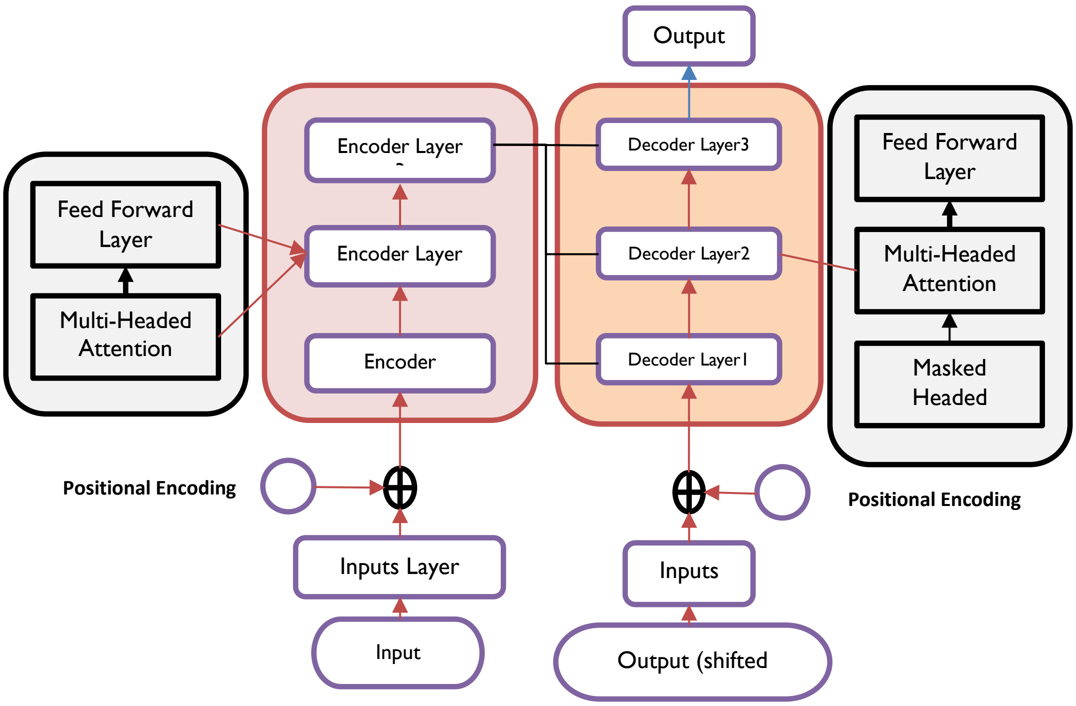


Figure 12: Detailed overview of the Transformer Model

5.4 Scaled Dot-Product Attention

The attention mechanism is carried out due to scaled dot-product mechanism. As mentioned initially, many studies have used the attention block for forecasting. What makes scaled dot-product attention different from all other is the introduction of the scaling factor $1/sqrt(d_k)$. With this introduction, the general attention mechanism equation is revised to the following:

$$Attention\ (Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \qquad (11)$$

Where d_k is the dimension of *K*.

Scaled Dot-Product Mechanism introduces three important matrices: Query (Q), Key (K), and Value (V). With the help of these matrices, or keys as they are called, the model is able to compute attention scores. These attention scores then help the model to concentrate on the right parameters and get the most accurate results out of them [3]. The attention scores are computed as the result of a dot product multiplication between the Query and Key vectors, as shown in the equation. The result of the multiplication is computed as a weighted sum of Values. A soft-max function is also put in the equation, which normalizes the result.

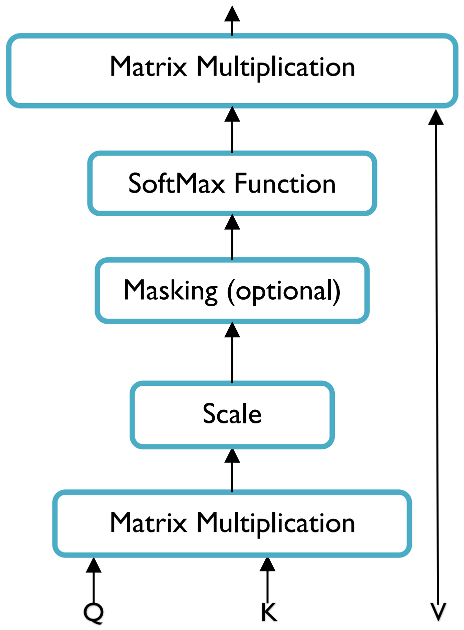


Figure 13: Scaled Dot-Product Multiplication Mechanism

5.5 Multi-Head Attention

The multi-headed attention block is defined as an extension of the self-attention mechanism. Multiple sets of the three matrices, each reflecting a different relationship between words, are used by multi-headed attention. A "head" of attention is the term used to describe each set. This enables the model to concentrate on many elements of the input sequence at once.

The ultimate output of the attention mechanism in multi-headed attention is created by concatenating and linearly transforming the outputs from each head. This enables the model to capture all kinds of dependencies and interactions between words in multiple embedding space subspaces [35].

The multi-headed attention block runs on the basic principle of self-attention. Each head in this model is operating as an independent self-attention mechanism model. The only difference here, in Wind Power Forecasting Model with a Natural Language Processing (NLP) Transformer, is that in our case, the transformer model is focusing on the power prediction parameters that will allow it to capture various patterns, and obtain visible results.

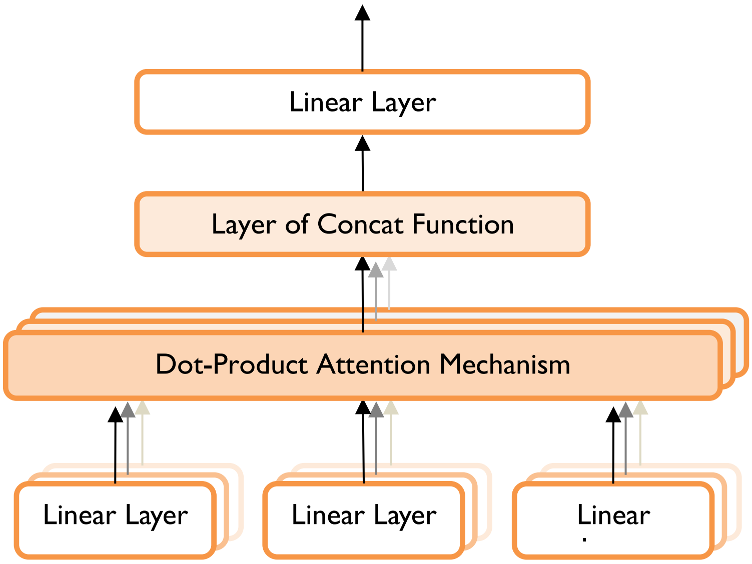


Figure 14: Multi-Headed attention mechanism

5.6 Position-wise Feed Forward Networks and Positional Encoding

Wind Power Forecasting requires that the location information of the time-series to be obtained at all cost. The Transformer model's ability to have parallel computation due to the presence of the Multi-Headed Attention block is a good benefit, but unlike CNN and other Recurrent Neural Networks, Transformer Model is unable to get the sequential information of the input of the time series pattern, i.e. it is not able to know at what time are we predicting an input [34]. This is where positional encoding comes to help the transformer model in doing this action. Better than using the absolute position of the wind power directly, the position encoding of each power value using trigonometric functions in a Transformer is well designed, utilizing its periodicity to model the position relationship between the inputs, and addressing its limitations of only being able to model sequences of finite length while taking advantage of the trigonometric function's value domain of [1,1], which can work well with input vectors [33]. The

Eq.(12) and Eq.(13) describe the general formula for positional encoding:

$$PE(\, pos, 2i) = \sin(pos/10000^{\overline{\frac{2i}{d_{model}}}} \qquad (12)$$

$$PE(\, pos, 2i + 1\,) = cos(pos/10000^{2i/d_{model}} \qquad (13)$$

With positional encoding, there is another layer which is connected to the encoder and decoder; the feed-forward networks [35]. Feed-forward networks consists of a a ReLU activation in between a linear transformation:

$$FFN(x) = max(\, 0, xW_1 + b_1)\, W_2 + b_2 \qquad (14)$$

Two linear transformations ensure that the input and output dimensions are identical to d_{model} while allowing for adjustment of the inner layer dimension.

6 Model Evaluation and Results

6.1 Model Evaluation

The evaluation employed Mean Absolute Percentage Error (MAPE), measuring model error magnitude and prediction accuracy. MAPE, a percentage counterpart of Mean Absolute Error (MAE), gauges error via Equation 15. MAPE suits cases with minimal outliers, values around zero, zeros, and sparse data. Despite being less outlier-sensitive than metrics like root mean squared error (RMSE), extreme low-end values can inflate MAPE. Pearson's coefficient of correlation (Equation 16) gauges actual-predicted value similarity for probabilistic forecasting.

$$MAPE = \frac{1}{n}\sum_{I}^{n} \left| \frac{A_i - F_i}{W_i} \right| \tag{15}$$

Where,

A_i is the actual reading

F_i is the forecast reading

W_i is the installed capacity of the wind turbine

n is the number of iterations

$$corr = \frac{\sum(xi-\bar{x})(y-\bar{y})}{\sqrt{\sum(xi-\bar{x})^2 \sum(yi-\bar{y})^2}} \tag{16}$$

Where,

x_i is the i^{th} sample of x

$\bar{x}$ = mean value of x-variable

y_i = is the i^{th} sample of y

$\bar{y}$ = mean value of y-variable

6.2 Data

Data used for this project was taken from the open-source hub operated by Sotvento wind farm situated in Galicia, Spain. Another dataset was acquired for this study via a wind farm in Turkey.

6.2.1 Turkey Wind Farm Dataset

An available historical SCADA dataset from Turkey is utilised in this investigation

[42]. Five parameters are taken from each record every ten minutes for a total of 52560 samples. The dataset included samples taken between January 1 and December 31, 2018.The dataset has some null fields, hence to fill those null values, linear interpolation is being employed. The following is a list of the dataset's parameters:

1. LV Active Power (KW);
2. Wind Speed (m/s);
3. Wind direction (°);
4. Theoretical Power Curve (kWh);

6.2.2 Sotavento Wind Farm Dataset

Data used for this project was taken from the open source hub operated by SOTAVENTO Wind Farm situated in Galicia, Spain. The location of the wind farm is near a river with on-shore wind turbines. The data is retrieved for every 10 minutes time stamp and up till two years therefore we had 105,120 values in total. The dataset included samples taken between November 1, 2020 and October 31, 2022 The variables used for the preprocessing were;

1. Wind angle
2. Wind speed
3. Power generated

And these three variables were used to predict the power generation.

6.2.3 Data Pre-processing

Parts of the dataset have been divided. 20% are utilized for testing, and the first 80% are used for training. The Turkey dataset contains 52560 samples hence its test dataset has 10512 data samples, whereas the train dataset has 42048 data samples. Whereas the sotavento dataset has 105,120 samples hence the testing dataset consist of 84,096 data samples while the testing dataset consist of 21,024 data samples. Both datasets contain some null fields and zero values which are then selected and eliminated from data and later linear interpolation technique is used to fulfill that data. Following steps were taken before the data into the model.

6.2.3.1 Normalization

In this step the data is normalized to a 0-1 scale using a min-max scaler:

$$X = \frac{\mathrm{X{-}Xmin}}{Xmax - Xmin} \quad (17)$$

Using the same parameters as the train dataset, i.e., the Xmin and Xmax test dataset, the train data is first normalised using this algorithm.

6.2.4 EEMD

The dataset is preprocessed and deconstructed for the use of EEMD. The findings demonstrate that although 14 IMFs were present in test data, 17 IMFs were discovered in train data. Now that every feature in the dataset has been broken down using EEMD decomposition, several IMFs have been generated for every parameter. The obtained IMFs and the residual portion are arranged from highest frequency to lowest frequency. For all parameters, the highest frequency component is removed from the original data because it represents the noisy portion of the dataset. To prevent the leakage of training data, this process was carried out independently for training and testing data.

6.2.5 CEEMDAN

The results with has EEMD improved a lot but there is a much better technique named as CEEMDAN. The process which was used for implementation of EEMD the same method is also repeated for the CEEMDAN and the results are quite satisfactory. The obtained IMFs and the residual portion are arranged from highest frequency to lowest frequency. For all parameters, the highest frequency component is removed from the original data because it represents the noisy portion of the dataset [43]. To prevent the leakage of training data, this process was carried out independently for training and testing data.

6.3. Parameter Selection for TCN and Transformer Models

6.3.1. TCN

For Temporal Convolution Network, the parameters were selected from [36] which are as follows:

Table 2: TCN Model Parameters

	Optimizer	**Number of filters**	**Kernal Size**	**Dilations**	**Number of Stack**
Parameter Value	Adam	32	10	[1,2,4,8,16]	2

6.3.2. Transformer Model

For the transformer model, we have added 3 layers of encoder-decoder mechanism within the model to allow parallel computation. The output dimension of the input layer, which is d_{model}, is set to 512. The number of multi-headed attention heads (MHA) are set to 8. The remaining hyperparameters for the transformer model were assumed, as we could not find enough sufficient material regarding them [38]. These hyperparameters were small in magnitude, which makes their overall effect on the model minimal.

6.3 Model Training results on Sotavento Wind Farm Dataset

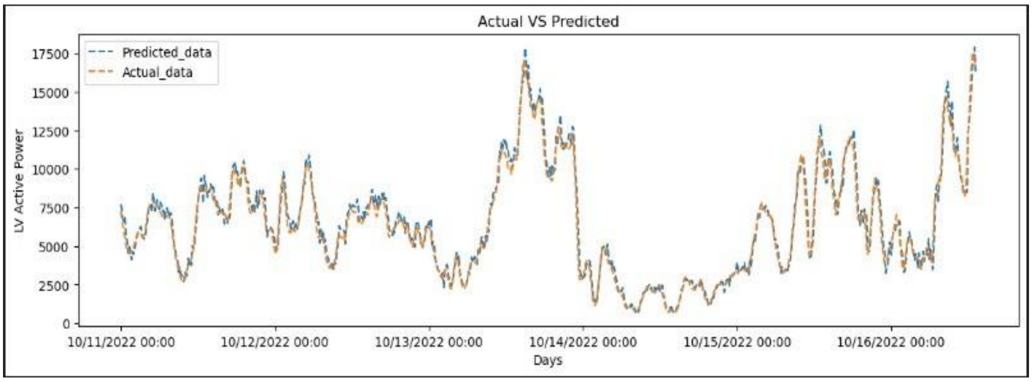


Figure 15: Raw TCN on Sotavento. The results shows that the accruacy of 98% and MAPE of 11%

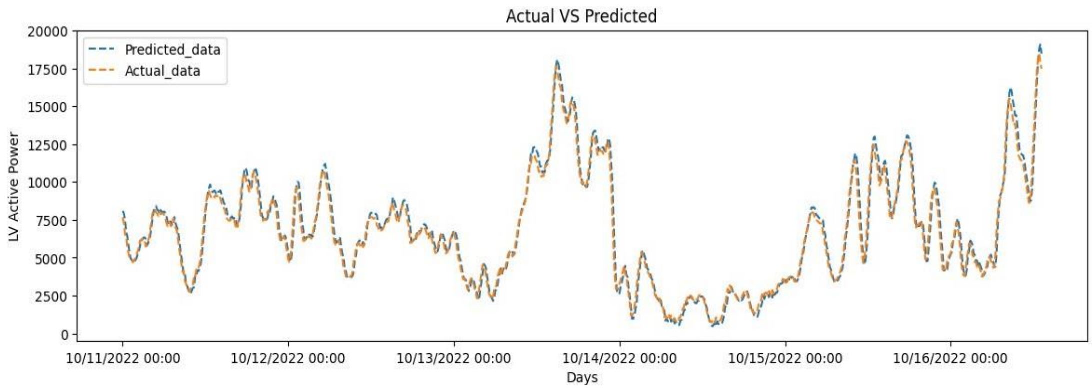


Figure 16: TCN with EEMD. The results shows that the accruacy of 99.30% and MAPE of 7.2%

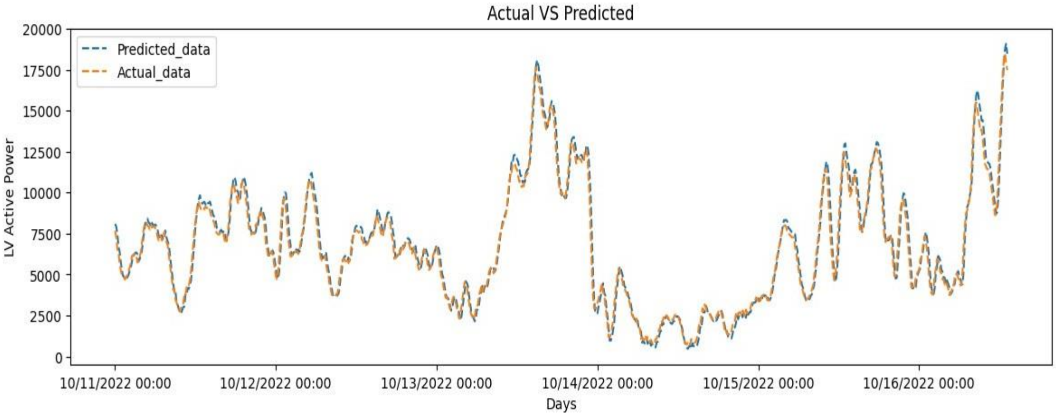


Figure 17: TCN with CEEMDAN. The results shows that the accruacy of 99.70% and MAPE of 5.2%

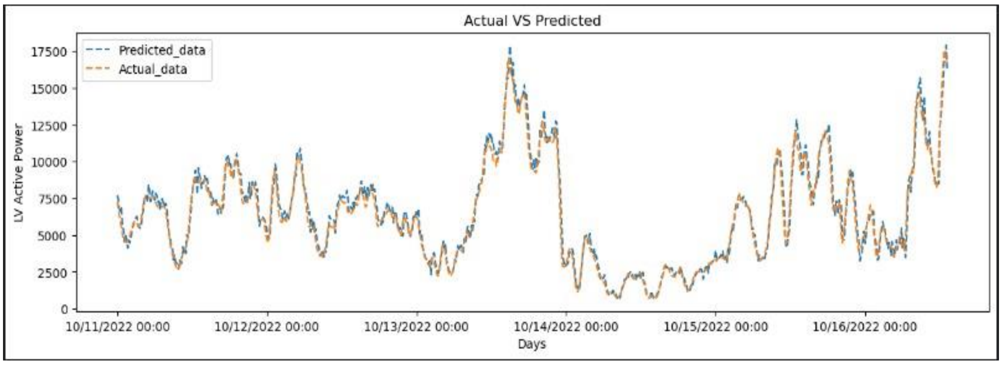


Figure 18: RAW Transformer. The results showed 97.7% accuracy and an MAPE of 8.6%

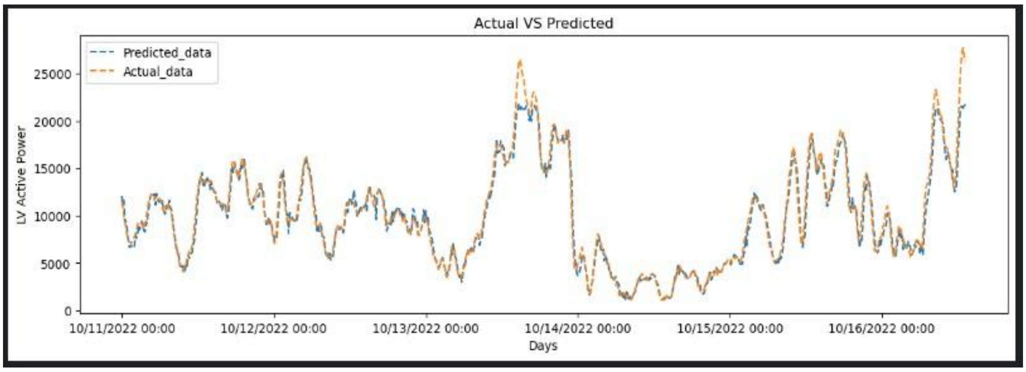


Figure 19: Transformer with EEMD. The results shows that the accruacy of 99% and an MAPE of 6.94%

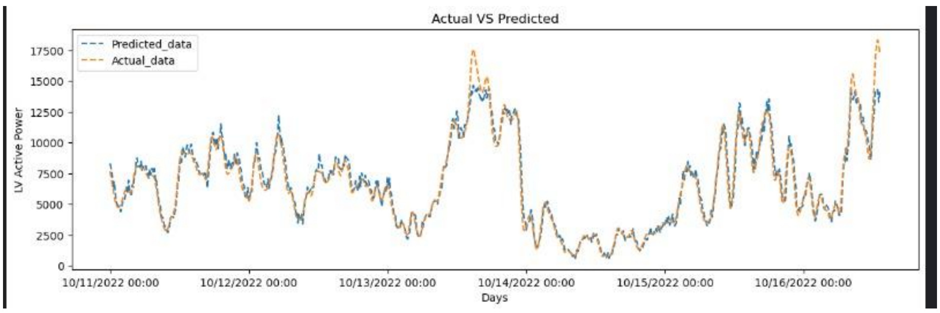


Figure 20:Transformer with CEEMDAN. The results shows that the accruacy of 99.2% and an MAPE of 6.7%

6.4.Model Training on Turkey Wind Farm Dataset

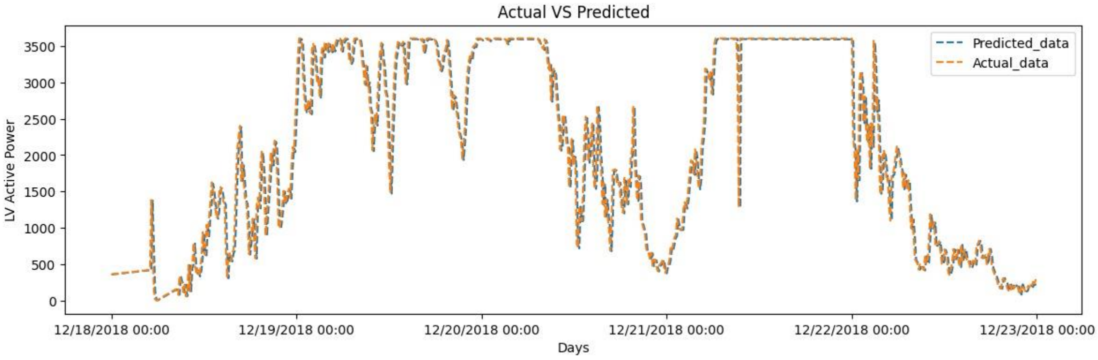


Figure 21: Raw TCN. The results show an accuracy of 97.30% and an MAPE of 14%

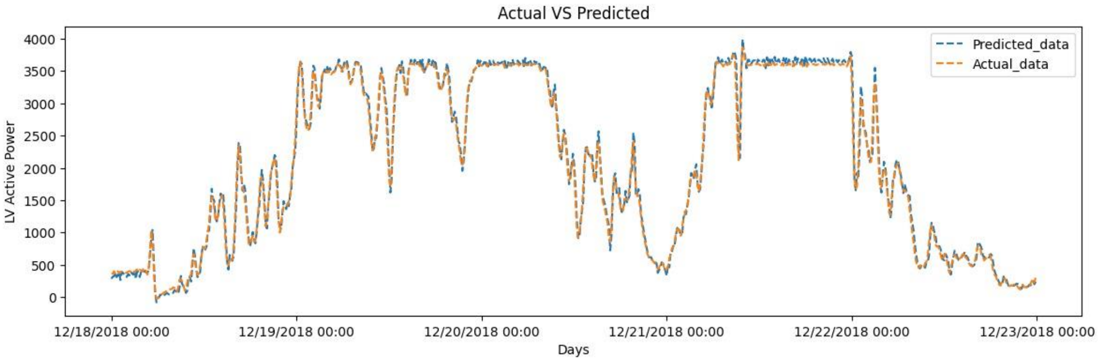


Figure 22: TCN with EEMD. The results showed an accuracy of 99.09% and an MAPE of 7.20%

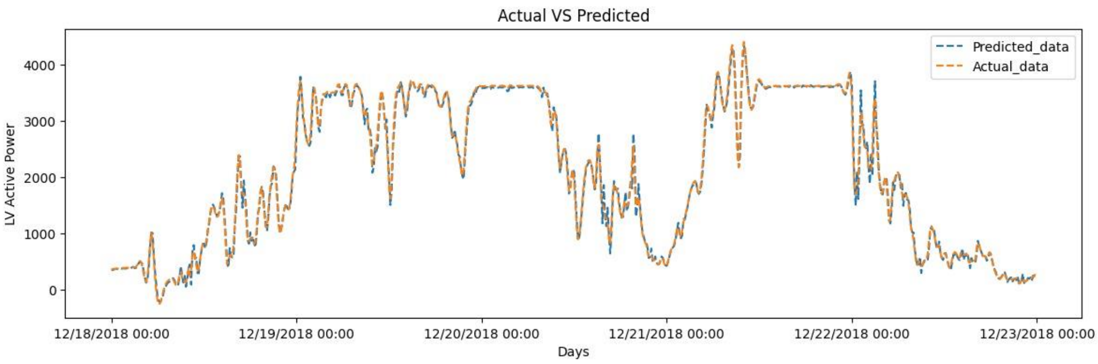


Figure 23: TCN with CEEDMAN. The results showed 99.7% accuracy and an MAPE of 5.4%

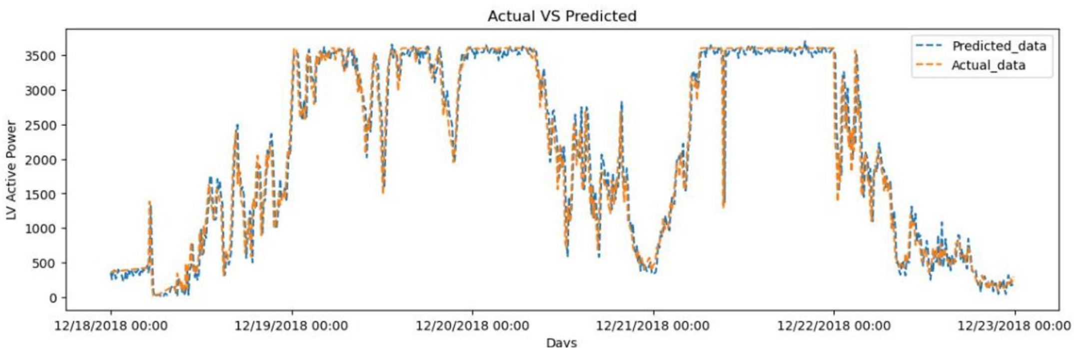


Figure 24: Raw Transformer. The results showed an accuracy of 97.7% and an MAPE of 16%

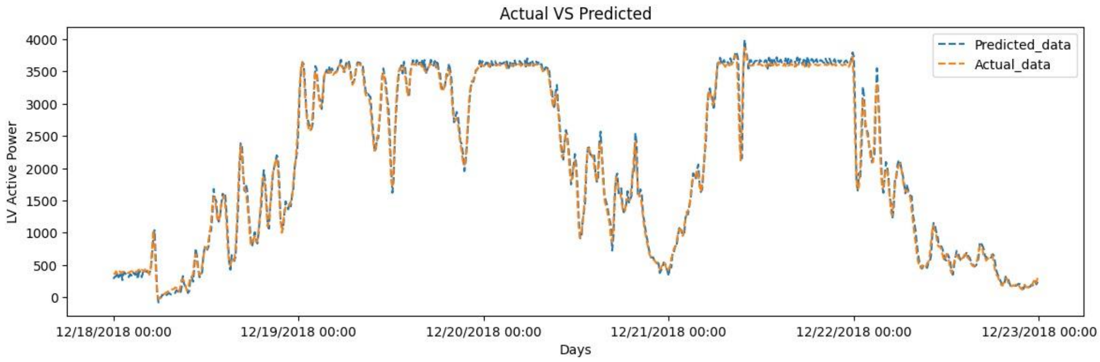


Figure 25: Transformer with EEMD. The results showed an accuracy of 99% and an MAPE of 12.5%

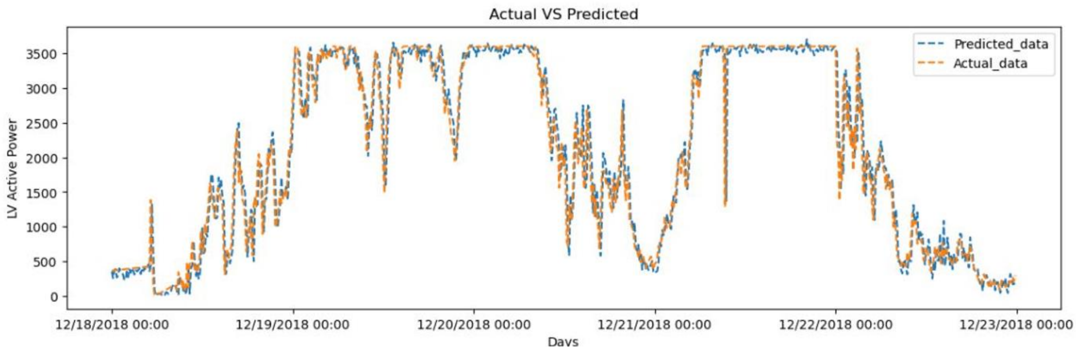


Figure 26: Transformer with CEEMDAN. The results show an accuracy of 99.2% and an MAPE of 12%

7 Conclusions

7.1 Summary

Working on two techniques; Temporal Convolution Network and Transformer Model to investigate the best predicting algorithm was our manifesto. But, during the research we came across the issue of erroneous data amalgamation, ultimately, resulting in false prediction and harming the accuracy. Different factors were analyzed and taken into consideration for the error detection but the previous studies and researches were not suggesting those solutions. Rather they were leading us to another field of research which was about the decomposition of signals.

However, we started working on eradicating the errors by working on TCN and Transformer techniques merger with Ensemble Empirical Mode Decomposition. This resulted in absolute increase in accuracy but we were not sure for the error percentage MAPE. We could see varying patterns in our graphs such as unnecessary peaks, unwanted null values and undesired wavelengths. To improve the variance in data, we used CEEMDAN which makes use of EMD and EMD for every IMF resulting in refined output. Thus, less noise and less errors are found in the graphs decreasing the percentage of error by significant numbers.

The tables show the evaluation of techniques along with decomposition techniques.

Table 3: Experimentation Results on Turkey Dataset

Model Training on Turkey Dataset				
Model Type	**TCN**		**Transformer**	
	MAPE	**Accuracy**	**MAPE**	**Accuracy**
Raw	14%	97.30%	16%	97.7%
With EEMD	7.01%	99.30%	12.50%	99%
With CEEMDAN	5.40%	99.70%	12%	99.2%

Table 4: Experimentation results on Sotavento Dataset

Model Training on Sotavento Dataset				
Model Type	**TCN**		**Transformer**	
	MAPE	**Accuracy**	**MAPE**	**Accuracy**
Raw	11%	98%	8.6%	98.18%

With EEMD	7.20%	99.30%	6.94%	98.58%
With CEEMDAN	5.2%	99.95%	6.7%	98.98%

7.2 Recommendations

This project was supposed to discover the wind power system. It was aimed to investigate the need and necessity of upgradation in this vary field. Usually, we find several researches and publications on power system management and fault analysis using ETAP and other softwares but it was not the same case with wind power sector. Wind sector is the biggest gem of renewable technology but it has always been neglected due to the unique requirements and government policies especially in the under-developed countries. Wind turbines and setup is expensive but economical for a long term because of its less maintenance and long life. Also, we require trained staff for this but the overall impact on environment, production of wind turbines, power generation and less area acquired by wind mills makes them favorable to use.

After the fruitful research on wind power forecasting, we have experienced that, wind energy can be very useful as compared to other forms of energies but there must be some effective mechanism to operate the systems. Our research proved that, integration of Artificial Intelligence into the power sector would definitely help the generation companies to utilize their resources in the right direction and yield the best outcomes.

Beside this, we have noticed that wind energy has particular trends throughout the year which should be analyzed and utilized for the maximum generation. Therefore, our recommendations would definitely go in the favor of wind power forecasting in comparison to solar residing the fact that you possess the ample resources.

8 References

[1] J. v. D. C. Klaiber, "Deep Learning for Variable Renewable Energy: A Systematic Review," *ACM Computing Surveys,* 2023.

[2] C. L. H. Gu, "Review on Deep Learning Research and Applications in Wind," *Energies,* 2022.

[3] S. Y. C. Q. Y. Huang, "Deep learning model-transformer based wind power forecasting approach," *Frontier,* 2023.

[4] Z. S. Y. X. Liu B., "A Novel Deep Learning Approach for Wind Power Forecasting Based on WD-LSTM Model," *Energies,* 2020.

[5] D. B. R. Koracin, "A REVIEW OF CHALLENGES IN ASSESSMENT AND

FORECASTING OF WIND ENERGY RESOURCES," *Environmental Science*, 2014.

[6] U. K. A. Datta, "The relevance of large-scale battery energy storage (BES) application in providing primary frequency control with increased wind energy penetration," *Journel of Energy Storage*, pp. 9-18, 2019.

[7] H. D. D. Dhiman, "A Review of Wind Speed and Wind Power Forecasting Techniques," *Electrical Engineering and Systems Designs*, 2020.

[8] S. N. N. Bosma, "Estimating Solar and Wind Power Production using Computer Vision Deep Learning Techniques on Weather Maps," *Electrical Engineering and Systems Science*, 2022.

[9] M. K. M. T. O. Noori, "A macro-level decision analysis of wind power as a solution for sustainable energy in the USA," *International Journal of Sustainable Energy*, 2015.

[10] M. G. E. Hussain, "Forecasting very short-term wind power generation using deep learning, optimization and data decomposition techniques," *24th International Conference on Electrical Machines and Systems (ICEMS*, 2021.

[11] T. K. V. A. M. R. K. Chou, "A Comeback of Wind Power in Shipping: An Economic and Operational Review on the Wind-Assisted Ship Propulsion Technology," *MDPI*, 2021.

[12] L. V. H. S. C. Heerdenn, "Wind Power Forecasting using Hybrid Recurrent Neural Networks with Empirical Mode Decomposition," in *2022 IEEE International Conference on Environment and Electrical Engineering and 2022 IEEE Industrial and Commercial Power Systems Europe*, 2022.

[13] J. Yan, "A Comparison Wind Power Forecasting with Different Methods," *Integrated Ferroelectrics*, 2022.

[14] Y. W. J. Liu, "Transfer learning based multi-layer extreme learning machine for probabilistic wind power forecasting," *Applied Energy*, 2022.

[15] C. Z. Y. Ding, "Integrated Carbon-Capture-Based Low-Carbon Economic Dispatch of Power Systems Based on EEMD-LSTM-SVR Wind Power Forecasting," *Energies*, 2022.

[16] J. Gonzalez-Sopena, "Ensemble Empirical Mode Decomposition Based Deep Learning Model for Short-Term Wind Power Forecasting," *Floating Offshore Energy Devices*, 2022.

[17] D. B. A. &. K. J. Nachimuthu, "Multi-step wind speed and wind power forecasting using variational momentum factor and deep learning based intelligent neural network models," *Concurrency and Computation*, p. 34, 2021.

[18] S. L. S. Z.-B. H. &. C. A. Hanifi, "Offshore Wind Power Forecasting—A New Hyperparameter Optimisation Algorithm for Deep Learning Models," *Energies*, 2022.

[19] H. N. D. Y. M. W. K. L. Y. &. X. X. Zhen, "A Hybrid Deep Learning Model and Comparison for Wind Power Forecasting Considering Temporal-Spatial Feature Extraction," *Sustainability*, 2020.

[20] N. &. L. Z. Wang, "A stacking-based short-term wind power forecasting method by CBLSTM and ensemble learning," *Journal of Renewable and Sustainable Energy*, 2022.

[21] S. Aggarwal and M. Gupta, "Wind Power Forecasting: A Review of Statistical

Models-Wind Power Forecasting," *International Journal of Energy Science*, 2013.

[22] J. P. V. &. G. B. Sopeña, "Decomposition-Based Hybrid Models for Very Short-Term Wind Power Forecasting," *Engineering Proceedings*, 2021.

[23] M. a. M. K. Ahmadi, "The linear-nonlinear data preprocessing based hybrid (LNDH) models for wind power forecasting," *Journal of Modelling in Management*, 2022.

[24] A. K. P. P. S. P. N. G. N. &. S. O. Rai, "Assessment of Different Deep Learning Models for Hardware Requirements and Pre-Processing Steps for Efficient Wind Power Forecasting," in *3rd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*, 2023.

[25] D. B. A. &. K. J. Nachimuthu, "Multi-step wind speed and wind power forecasting using variational momentum factor and deep learning based intelligent neural network models," 2021.

[26] Y. M. L. N. W. &. Y. C. Zhou, "Data Enrichment as a Method of Data Preprocessing to Enhance Short-Term Wind Power Forecasting," *Energies*, 2023.

[27] J. S. J. &. R.-M. M. Maldonado-Correa, "Wind power forecasting: A systematic literature review," *Wind Engineering*, 2019.

[28] J. C. Y.-R. Tang, "Research on Wind Power Short-Term Forecasting Method Based on Temporal Convolutional Neural Network and Variational Modal Decomposition," 2022.

[29] H. D. X. S. R. &. W. S. Luo, "A Multi-Step Prediction Method for Wind Power Based on Improved TCN to Correct Cumulative Error," *Engineering*, 2021.

[30] J. S. L. &. L. Y. Zhu, "Wind power forecasting based on new hybrid model with TCN residual modification," *Energy and AI*, 2022.

[31] X. Z. L. G. H. &. L. X. Jiang, "A Temporal Convolution Memory Network Model for Short-Term Wind Speed Forecasting," *2023 8th Asia Conference on Power and Electrical Engineering (ACPEE)*, 2023.

[32] C. Z. Y. D. Q. &. L. K. Ding, "Integrated Carbon-Capture-Based Low-Carbon Economic Dispatch of Power Systems Based on EEMD-LSTM-SVR Wind Power Forecasting," *Energies*, 2022.

[33] M. H. M. &. S. S. Faisal Aftab, "Detecting non-linearity induced oscillations via the dyadic filter bank property of multivariate empirical mode decomposition," *Journal of Process Control*, pp. 68-81, 2017.

[34] S. M. G. Devi, "Hourly day-ahead wind power forecasting with the EEMD-CSO-LSTM-EFG deep learning technique," *Soft Computing*, 2020.

[35] Z. &. H. N. Wu, "Ensemble Empirical Mode Decomposition: a Noise-Assisted Data Analysis Method," *Advances in Adaptive Data Analysis.*

[36] H. &. B. S. You, "New PSO-SVM Short-Term Wind Power Forecasting Algorithm Based on the CEEMDAN Model," *J. Electr. Compute. Eng.*, 2022.

[37] M. &. G. A. Maurya, "Attention Based Short Term Wind Power Forecasting Using Sample Entropy And Correlation Analysis for Noise filtration," in *IEEE International Conference on Power Electronics, Smart Grid, and Renewable Energy (PESGRE)*, 2022.

[38] S. Y. C. Huang, "Deep learning model-transformer based wind power forecasting approach," *Frontiers*, 2023.

[39] Y. Z. T. W. W. Q. R. M. Niu Z., "Wind power forecasting using attention-based gated recurrent unit network," *Energy,* 2020.

[40] S. Z. K. X. Y. X. Qu K., "Short-term forecasting for multiple wind farms based on transformer," in *The 2nd International Conference on Power Engineering (ICPE 2021),* Guangxi, 2021.

[41] N. S. N. P. J. U. L. J. A. N. G. Ł. K. I. P. Ashish V., "Attention is All you Need," in *Advances in Neural Information Processing Systems 30* , 2017.

[42] W.-H. Lin, P. Wang, K.-M. Chao, H.-C. Lin, Z.-Y. Yang and Y.-H. Lai, "Wind Power Forecasting with Deep Learning Networks: Time-Series Forecasting," *Applied Sciences,* 2021.

[43] D. C. J. Yu, "Application of EMD method and Hilbert spectrum to the," *Mechanical Systems and Signal Processing,* pp. 259-270, 2005.

[44] W. W. P. C. K. &. L. H. Lin, "Wind Power Forecasting with Deep Learning Networks:," *Applied Sciences,* 2021.

[45] H. N. D. Y. M. W. K. L. Y. &. X. X. Zhen, "A Hybrid Deep Learning Model and Comparison for Wind Power Forecasting Considering Temporal-Spatial Feature Extraction," *Sustainability,* 2020.

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Publication

8.1.5 www.meca.unicaen.fr
Internet Source

8.1.6 J. Benajes, R. Novella, J.M. Pastor, A. Hernandez-Lopez, T. Duverger. "A computational analysis of the impact of bore- to-stroke ratio on emissions and efficiency of a HSDI engine", Applied Energy, 2017

Publication

8.1.7 digital.library.adelaide.edu.au

Air or in Diesel Emulsion on the Performance and Emissions of a HDDI Diesel Engine", journal of Energy Engineering, 2020

Publication

---

11 N. Saravanan. "Hydrogen-diesel dual fuel combustion in a direct injection diesel engine", International journal of Renewable Energy Technology, 2011 <1 %

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